



4D STEM

Chapter Preview

In STEM experiments, conventional detectors measure only a few intensity values at each probe position, using monolithic single-pixel detectors. However, as we have seen in previous chapters, DPs are extremely rich in structural information about our specimen—can we record this information at each STEM probe position?

With new electron detector technology, we can record thousands or even hundreds of thousands of images per second, allowing us to measure DPs at typical STEM scanning speeds (tens of microseconds dwell time per probe position). A dataset containing full 2D DPs over a 2D grid of probe positions is four dimensional, giving us the name ‘4D-STEM’. These experiments produce a prodigious amount of data, far more than typical 2D images or 3D experiments such as spectroscopy or tomographic imaging. We therefore need to employ fast and robust computational methods to analyze these datasets. In this chapter, you’ll learn the theory and mathematical basis of many of these analysis methods.

As you’ll see, 4D-STEM is a very broad umbrella, which encompasses massively repeated conventional diffraction experiments with nearly parallel illumination, nanobeam electron diffraction measurements of local structure, and advanced phase contrast diffractive imaging methods which push the bounds of imaging resolution. These methods are all possible because STEM is a very flexible operating mode for a TEM. We can tune the probe size by orders of magnitude, and our measured DPs contain STEM probe momentum distributions over a wide range of scattering angles. Many of these experiments produce different types of data for ordered and disordered specimens, and therefore require specimen-specific analysis methods.

By the end of this chapter, hopefully you will agree that 4D-STEM experiments are worth the additional experimental and data analysis challenges. We expect that most STEM experiments will be 4D-STEM in the near future, so you might as well learn now!

17.1 What Is 4D-STEM?

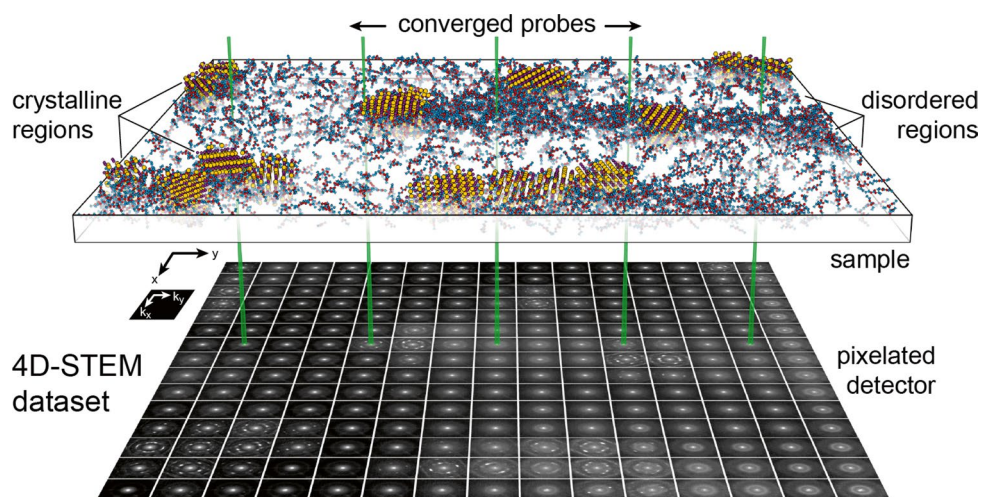
This chapter describes a family of techniques which all fall under the banner of 4D-STEM. These experimental methods use a pixelated detector to record a 2D image of the STEM probe after passing through the specimen, usually recorded over a 2D grid of probe positions. The resulting dataset is four dimensional, hence the term 4D-STEM. This name is quite general by design, as you'll see that it covers a fair number of different techniques.

The most common 4D-STEM configuration we use is to record the 2D images of the STEM probe in diffraction space; thus, most 4D-STEM experiments can be thought of as 'massively repeated' diffraction experiments, with many of the same strengths and weaknesses as the diffraction methods covered in W&C ► Chaps. 18, 19, 20 and 21 and ► Chaps. 7, 8 and 9 in **this Companion Volume**. In ► Fig. 17.1 you can see a schematic of a 4D-STEM experiment and a high-level representation of example 4D dataset collected from a specimen composed of a complex mixture of ordered (crystalline) and disordered phases.

We use various names for specific 4D-STEM techniques, and these are classified by several variables: the probe-convergence angle (α —also known as the numerical aperture), the incident-probe wavefunction shape, whether the probe measurement is performed in diffraction space or real space, and the specimen properties we are measuring. ► Table 17.1 gives the name of and summarizes many of these techniques, their range of operating parameters, and some possible applications.

While we've attempted to make the Table as complete as possible, unfortunately, many of these methods overlap each other and because the techniques are relatively new, there is often no universally agreed upon definition of them. This situation is unlikely to improve as the number of 4D-STEM studies published in the literature is rapidly expanding. The term '4D-STEM' has proven to be relatively sticky, and coupled with its generality, we expect it will continue to find widespread use. Sometime in the future however, we predict that collecting 4D-STEM datasets will be the default for all STEM imaging experiments and thus these methods may be just called STEM or even just another TEM class of techniques.

By a large margin, the most important experimental parameter for a 4D-STEM experiment is the maximum scattering angle of your probe-forming aperture, which is typically a circle. (Remember, as you learned in ► Chap. 16, this probe-forming aperture is correctly termed the objective aperture but it is actually the C2 aperture in the microscope illumination system.) This angle is usually measured from the optical axis to the aperture edge and thus defines the 'radius' of the STEM probe. Note that since it covers only half of the probe diameter, we usually refer to it as a 'semi-angle' and we'll denote this as α_{probe} (see ► Chap. 6 of W&C). The probe's convergence angle usually means the full angular range of the probe, or the 'diameter' of the STEM probe. The primary reason why this single parameter is so important is that it sets the minimum size for our probe in the specimen plane in real space, and thus the smallest length scales which can be probed in the experiment. If we wish to see atomic resolution, our probe needs to be smaller than the atomic spacing, or it will blur the resulting measurement too much to resolve atoms (or atomic planes).



► **Fig. 17.1** Example 4D-STEM experiment. A converged STEM probe is scanned in a 2D grid over a complex specimen containing both ordered crystalline and disordered regions. At each probe position, we collect a full 2D DP, producing a 4D dataset. Adapted from Ophus (2019) and Xie (2023)

■ **Table 17.1** Non-exhaustive list of 4D-STEM and adjacent methods

Acronym/initials	Full name	Application comments
4D-STEM	Four-dimensional scanning transmission electron microscopy	General term for all experiments where the recorded data has dimensionality 4 or higher
CBED	convergent beam electron diffraction	Common name for all experiments using a converged beam, diffracted disks may or may not be overlapping
NBED	Nanobeam electron diffraction	Common name for experiments where the probe convergence semi-angle is 0.1–5 mrad (non-overlapping diffraction disks)
SEND	Scanning electron nanodiffraction	Common name for experiments where the probe convergence semi-angle is 0.1–5 mrad (non-overlapping diffraction disks)
PRD	Position-resolved diffraction	Less common name for SEND or NBED
SRD	Spatially resolved diffractometry	Less common name for SEND or NBED
MR-STEM	Momentum-resolved STEM	Less common name for SEND or NBED
PED	Precession electron diffraction	Beam is precessed (tilted around a cone of angles from the optic axis), with the aim of reducing strong diffraction contrast, thus making diffraction more reliable and interpretable
4D-SPED	Four-dimensional scanning PED	Scanning PED experiments collecting where the full 4D dataset is recorded
DPC	Differential phase contrast	Generic name for experiments which measure the deflection the probe due to either the specimen structure, electric fields, or magnetic fields
CoM	Center-of-mass	Average momentum vector of the diffraction probe measured with a pixelated detector to measure phase gradients, which can then be integrated to recover object phase
	Ptychography	Iterative or non-iterative deconvolution of the probe and object wavefunctions, typically applied at high- or atomic-resolution
	Microdiffraction, nanodiffraction, picodiffraction	Alternative names for DPs measured from probes with varying degrees of convergence
	Ronchigram, diffractogram	Intensity of the diffracted probe, typically of an amorphous specimen to characterize the probe aberrations
	Pixelated STEM, universal detector	Non-monolithic detectors used in 4D-STEM experiments
ACOM	Automated crystal orientation mapping	Automatic indexing of Bragg DPs from a diffraction library to determine the local phase(s) and crystal orientation(s)
PDF/RDF	Pair/radial distribution function	Estimated distribution of bonds (2-body distribution) in real space, modulated by atomic form factor. Probes short-range order (SRO)
FEM	Fluctuation electron microscopy	Estimates the degree of disorder from “speckle” DPs by measuring the diffracted intensity variance as a function of scattering angle. Also considered a probe of medium-range order (MRO)
STEMH	STEM holography	The name of an older technique where a biprism is used in STEM (see §17.3.4 for more information) and a newer method where a diffraction grating splits the STEM probe into multiple beams, which pass through the specimen and recombine at the detector
4D-SCED	Four-dimensional scanning confocal electron diffraction	Specimen moved out of cross-over, beams are focused into a DP using a confocal configuration. The intersection of the probe with the specimen in 4D-SCED is a hard-edge circle (no probe tails)
4D-SCEM	Four-dimensional scanning confocal electron microscopy	By combining both a probe and image corrector, the real space intensity distribution of the STEM probe can be imaged in real space after interacting with a specimen. Allows for depth sectioning
R-STEM	Real space STEM	Image of the probe is recorded in real space (conjugate plane to the specimen), typically as high or atomic resolution. Alternate name for 4D-SCEM

Other key parameters that we can control in a 4D-STEM experiment are the accelerating voltage, the amount of defocus and other wave aberrations such as Cs included in the STEM probe, the range of angles we measure on our pixelated detector (set by the camera length), and the step size between adjacent measurements. If we change the accelerating voltage then, as you know by now, higher voltage producing higher resolution images but often with specimen damage tradeoffs (which we explored back in ► Chap. 5).

17.2 Motivation: Why Collect So Much Data?

In a conventional STEM imaging experiment, we focus the beam to the desired resolution (probe size) and then scan it over our specimen surface, usually in a 2D grid of equally spaced probe positions (covered in detail in ► Chap. 16). We then use a small number of large, monolithic and single-pixel detectors to record the electrons over a range of scattering angles. This is fundamentally an information-destroying process which ultimately limits the information we can retrieve from our specimen. If we collect all electrons less than a maximum α , we call this experiment a BF measurement. This is because the background signal (when the probe is passing through vacuum) will be as high as possible, leading to a bright background signal; hence the name. Note that if we collect electrons only up to a maximum angle much smaller than the value of α , this experiment is still called BF, and will produce contrast very similar to plane-wave BF TEM measurements due to the reciprocity theorem (see §22.6 in W&C). A detector configuration which has both a minimum and maximum scattering angle (typically 1:2 ratio) where the maximum angle is equal to the probe's maximum scattering angle is called ABF.

We will now summarize in a paragraph what you have read in ► Chaps. 15 and 16 because it is really important, we are in trouble if this is not totally clear, and we know some of you will not read the chapters in order!

If we collect all electrons greater than a minimum semi-angle which is larger than the probe's convergence semi-angle, we refer to this as an ADF experiment. These experiments can be further divided up into low angle LAADF, medium-angle (MAADF), and of course high-angle (HAADF). The distinction we make between LAADF and MAADF is fuzzy, but both typically include both coherent diffraction and (more) incoherent thermal-diffuse scattering in different ratios. HAADF however is usually defined as using a minimum detector angle such that we record *no* coherent diffraction, producing what is typically referred to

as a Z-contrast image. HAADF is (by a large margin) the most popular STEM imaging method (as you'll have gathered from ► Chaps. 15 and 16) in materials science, because we can easily interpret (and model) such images, and they contain reasonable elemental sensitivity. However, HAADF imaging has substantial weaknesses, as it produces little or no contrast from weakly-scattering elements such as O, C and other light elements, and has poor dose efficiency (we need many electrons to image specimens at high resolution, requiring an electron dose higher than some specimens can tolerate).

Segmented-detector DPC experiments typically use 4–16 monolithic single-pixel detectors, but rather than being radially symmetric like traditional BF or ADF detectors, these detectors are usually arranged in quadrants. (While this detector design is not conceptually new (see Fig. 29.5 in W&C) recent improvements in detectors have been dramatic.) Quadrant detectors allow a DPC measurement to estimate the deflection of the probe, by measuring the *difference* between opposite detectors (e.g. left minus right or top minus bottom). This allows us to perform more dose-efficient, phase-contrast measurements. As we'll see below in §17.6.1, we can integrate these DPC signals to estimate the phase shift imparted by the specimen or differentiate these signals to estimate the charge density. With quadrant-detector DPC measurements, we must be careful to align the microscope such that the probe does not shift in the detector plane as we scan it over our specimen (i.e., we must minimize beam de-scan).

The first reason we perform 4D-STEM experiments is to overcome a substantial weakness of the conventional measurements that we listed above; they each require a fixed detector configuration, which is set prior to collecting data. If we instead use a pixelated detector, we can define our detectors *after* the experiment using software and optimize the resulting contrast. This is why 4D-STEM measurements are sometimes referred to as using a 'universal detector' since, if we choose a camera length such that we capture the electrons scattered to angles as high as is reasonably probable from a given specimen, then we can, in theory, construct any detector configuration we wish. We can also use multiple overlapping detectors, which is not possible with physical detectors without subdividing each unique geometric region. By contrast, in a conventional experiment, each time we want to try another detector configuration we need to record another dataset.

The second reason we perform 4D-STEM experiments is because we can use computational imaging to enhance many of our measurements using software. We

can do this live while we are on the microscope (though usually only for simple, non-iterative computational methods) or as a post-processing step. For example, rather than adding up electrons in each scattering range in the manner of a conventional detector, we could add some regions with different weights, subtract others using negative weights, and ignore the remaining angular ranges with zero-weights. We can use simulation to determine these special detectors, or methods such as principal-component analysis (PCA—much more about this in ► Chap. 23) on the full 4D-STEM dataset to compute virtual detectors which maximize the contrast from the specimen. Or for a DPC measurement, we can compute the center-of-mass (CoM) (first moments) of each detector pattern, instead of performing two difference measurements from pairs of detectors on opposite sides of the optic axis. This removes the directional anisotropy of DPC from quadrant detectors, where we are more sensitive to probe shifts in the up-down and left-right directions than the diagonals. We can also push beyond the resolutions of DPC by computationally solving for the STEM probe wavefunction and deconvolving it from the specimen, using ptychography (introduced in ► Chap. 16 and expanded later in this chapter). DPC measurements will be blurred by the shape of the probe which, if it contains substantial defocus, astigmatism, or other aberrations could represent a substantial loss of resolution.

The third reason you should think about performing 4D-STEM measurements is to access information which you can't obtain in a conventional STEM or TEM experiment. For example, a crystalline specimen will produce Bragg diffraction spots which encode the projected structure factor of the material, and you can use this information to determine the crystal phase, orientation, and local deformation of your specimen. In a conventional BF-, ADF-, or DPC-STEM imaging experiment, we could perhaps determine which regions of the specimen diffract more or less electrons and measure the resulting grain shapes and sizes, but the specific DPs are lost. In a conventional TEM SAD (see ► Chap. 7) experiment, we could measure these DPs, but depending on the size of our aperture we might mix several or even hundreds of DPs from differently oriented crystalline grains together. With 4D-STEM, we don't need to make these tradeoffs; if we scan a crystalline specimen with a nanometer-scale probe, we can collect the full DP at each probe position and record the resulting Bragg diffraction. We can also easily separate out radially isotropic signals such as the 'amorphous halo' from amorphous or disordered specimen regions from the sharper Bragg diffraction spots produced if we have crystalline specimen regions. We can also perform multiple analyses or experiments from

each dataset—such as interphase orientation determinations—which are the focus of ► Chap. 7.

The final reason for us to perform 4D-STEM experiments, and the one we believe is most important for quantitative characterization of material properties, is that we can fully decouple the resolution of our measurements from the field of view. If we perform a HR imaging experiment, we need to use a small enough step size between adjacent STEM probes to resolve the features of interest (we must step more finely than the Nyquist limit). This sampling requirement strongly limits our achievable field of view. Often the functional length scale of a complex specimen ranges from hundreds of nanometers up to hundreds of micrometers, while the features of interest are nanometer- or atomic-scale. Additionally, we may wish to record a statistically significant number of specimens, for example the crystal structure or orientation of thousands of nanoparticles. This could require us to take many atomic-resolution images—or only a single 4D-STEM experiment.

17.3 Experimental Details

17.3.1 Detectors for 4D-STEM

Most 4D-STEM hardware instrumentation is the same as we use in routine STEM imaging experiments, as we described in ► Chap. 16. However, as we just discussed above, to perform a 4D-STEM experiment we need to replace our monolithic single-pixel detectors with a pixelated imaging detector. A conventional imaging detector (check back in W&C ► Chap. 7) consists of a CCD, with a scintillator on top to convert incoming electrons into a shower of photons. These detectors typically have a poor PSF, meaning that a single electron strike can be blurred out into several adjacent pixels. Additionally, these detectors may suffer from poor detective quantum efficiency (DQE), meaning that many incident electrons are not measured. Additionally, because the scintillator CCD usually produces an 'afterglow', it has a limited time resolution, often no more than video frame rate (10–100 frames/s). A CCD detector operating at 50 frames/s can record one STEM probe in 20 ms. This is far longer than a typical STEM probe dwell time, which ranges from 1 to 20 ms. Consequently, CCDs are poorly suited to 4D-STEM experiments.

What we primarily need for a good 4D-STEM experiment is the addition of a direct electron detector (DED). In contrast to a CCD, a DED does not require

a scintillator and instead directly measures the incoming electron signal. This means that DEDs can have far better performance characteristics. These advantages include a much higher DQE for each electron, less blurring of the measured electron, a more linear signal over a large range of electron densities (i.e. dynamic range), and much higher frame rates (all of which comes at a price!). This last property is especially important for 4D-STEM; for example, if we were to scan 300×300 probe positions using a video-rate CCD (30 frames/s), it would require 50 min to complete. That same scan performed with a DED operating at 30,000 frames/s would require only 3 s to complete! Thus, detectors which can operate at high speed are a key enabling technology for 4D-STEM. Many of these detectors produce >1 GB/s of data, which is why we require fast computational methods when we're doing 4D-STEM analysis.

Currently we use two classes of detectors in 4D-STEM experiments. The first is active pixel sensors (APS), which are CMOS chips which are doped with an electron-sensitive layer. These detectors offer very high speeds and large numbers of pixels (up to tens of millions), but these advantages are offset by a poor dynamic range. The second class are hybrid pixel array detectors (PADs). These detectors use an array of photodiodes, with complex readout electronics behind each pixel. These detectors possess a very high dynamic range and high readout speeds but are limited to a small number of pixels. Typically, we prefer PADs for measurements such as ptychography where we require high dynamic range, while APS detectors are better suited to nanobeam-diffraction experiments where a higher number of pixels can improve the measurement accuracy.

A 4D-STEM measurement is four dimensional, with two real space and two diffraction (reciprocal) space dimensions. This data is typically stored as a numerical matrix

$$I(\mathbf{R}, \mathbf{K}) = I(R_x, R_y, K_x, K_y), \quad (17.1)$$

where $\mathbf{R} = (R_x, R_y)$ are the real-space coordinates of the STEM probe, and $\mathbf{K} = (K_x, K_y)$ are the diffraction-space coordinates on the detector. It's important that we store the data in the correct order on our disk drives, as we typically want each DP to be contiguous (in continuous order) in the data file, so that we can analyze images sequentially if needed. The dimension ordering given above is the one used in programming languages such as C and Python.

17.3.2 Probe Convergence Semi-Angle

The most important experimental parameter for most STEM experiments is the size of the probe-forming aperture. As you know (from way back in ► Chap. 9 of W&C) this aperture size is usually given as a radius and it defines α_{probe} . It sets the maximum tilt angle of electrons away from the optical axis which form the STEM probe, which we can geometrically think of as having the shape of a cone. The maximum tilt can be defined in terms of scattering angle α , or using the scattering given by the reciprocal space coordinate $\mathbf{k} = (k_x, k_y)$. These variables are linearly related by the (relativistic) electron wavelength λ , by the expression $\alpha_{\text{probe}} = \lambda k_{\text{probe}}$. In light and X-ray optics, this parameter is often given as the numerical aperture (NA), equal to $\text{NA} = \sin(\alpha_{\text{probe}})$, which for the small angles used in TEM is essentially equal to α_{probe} . Mathematically, we can define an initial STEM-probe wavefunction $\Psi_0(\mathbf{k})$ using the expression.

$$\Psi_0(\mathbf{k}) = \begin{cases} 1, & |\mathbf{k}| \leq k_{\text{probe}} \\ 0, & |\mathbf{k}| > k_{\text{probe}} \end{cases}. \quad (17.2)$$

This expression represents an ideal, aberration-free probe. Because we form the probe in reciprocal (diffraction) space, a larger value of k_{probe} leads to a smaller probe in real space on the specimen surface, and vice versa. We describe an ideal probe in terms of the real space coordinates $\mathbf{r} = (r_x, r_y)$ by the equation.

$$\psi_0(\mathbf{r}) = \frac{J_1(2\pi|\mathbf{r}|k_{\text{probe}})}{\sqrt{\pi}|\mathbf{r}|}, \quad (17.3)$$

where $\psi_0(\mathbf{r})$ is equal to $\psi_0(\mathbf{r}) = \mathcal{F}_{\mathbf{k} \rightarrow \mathbf{r}}\{\Psi_0(\mathbf{k})\}$, i.e. the inverse FT of the diffraction-space probe, and J_1 is the Bessel function of the first kind. The relative number of electrons scales with the probe intensity, given by the absolute magnitude squared as $I(\mathbf{r}) = |\psi_0(\mathbf{r})|^2$. Equation 17.3 has an infinite number of zero crossings, and therefore an in-focus STEM probe consists of a central lobe and many concentric rings with decreasing intensity. Approximately 87% of the probe intensity is contained within the central lobe, with an intensity distribution approximately equal to

$$I_{\text{center}}(\mathbf{r}) \approx |\cos(2.6|\mathbf{r}|k_{\text{probe}})|^{2.8} \text{ where } r < 0.61/k_{\text{probe}}, \quad (17.4)$$

The reason α_{probe} is so important is that it sets the minimum size of the probe on the specimen surface, and the degree of overlap in the DPs we measure in 4D-STEM. As we'll see in the sections below, different 4D-STEM

experiments require quite different values for α_{probe} , ranging from $\alpha_{\text{probe}} = 0.1$ mrad for pair-distribution function mapping, to 0.5–5 mrad for nanobeam electron diffraction (NBED), to >10 mrad for atomic-resolution virtual imaging or ptychography, and >30 mrad for depth-sectioning experiments. Most modern microscopes come equipped with a range of different sizes of circular probe-forming apertures in their condenser systems. As we'll discuss in §17.3.4, we can also fabricate custom apertures.

17.3.3 Probe-Aberration Function

All real optical systems have aberrations. You know this is especially true for electron-optical systems, as we cannot make round electron lenses without Cs, a fact proven by Otto Scherzer back in 1936 (and discussed many times in both W&C and chapters in this volume). With modern aberration correctors, we can reduce or even eliminate Cs and other low-order aberrations. However, these correctors will introduce higher-order aberrations, especially those which vary in the annular direction. We define an aberration function $\chi(\mathbf{k})$, which includes all aplanatic aberrations, meaning aberrations which depend only on the scattering vector. We can incorporate this aberration function into the initial STEM probe using the expression.

$$\Psi_0(\mathbf{k}) = \begin{cases} \exp[-i\chi(\mathbf{k})], & |\mathbf{k}| \leq k_{\text{probe}} \\ 0, & |\mathbf{k}| > k_{\text{probe}} \end{cases} \quad (17.5)$$

There is one commonly used set of basis functions that we use to define aberrations of different orders, but there are various conventions for the pre-factors. For convenience, here we define aberrations using dimensionless pre-factors as

$$\chi(\mathbf{k}) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} |\alpha|^{2m+n} \left\{ C_x^{m,n} \cos(n\phi) + C_y^{m,n} \sin(n\phi) \right\}, \quad (17.6)$$

where m is the radial order, n is the annular order, $(C_x^{m,n}, C_y^{m,n})$ are the cartesian components of the aberration coefficients for a given order (m, n) , and $\phi = \arctan 2(k_y, k_x)$ is the angular direction. You can see that the radially symmetric aberrations ($n = 0$) have only a single aberration coefficient $C^{m,0}$.

STEM probe aberrations do not typically affect 4D-STEM experiments with a small $\alpha_{\text{probe}} (< 10 \text{ mrad})$. However, when we perform atomic-resolution 4D-STEM measurements such as those we'll describe in §17.6 often the resolution is limited by aberrations, or by how effectively we can measure and correct the residual aberrations.

17.3.4 Amplitude and Phase Plates

The STEM probes we just described use circular probe-forming apertures. We can however use more complex apertures or even other electron-optical elements to form STEM probes. For example, in NBED experiments, we typically use amplitude plates to impart a pattern into the intensity of the probe and therefore the DP, to improve measurement precision and accuracy. They have also been used for more exotic 4D-STEM measurements, including forming multiple (incoherent) STEM beams which impact the specimen at different angles, or phase measurements which use hard edges to directly recover specimen structure from the DPs. Phase plates have an even larger variety of potential uses, including aberration correction, probe-wavefunction control, splitting the probe into multiple (coherent) beams in STEM holography, direct measurement of specimen structures using interferometric scattering, and adding phase diversity into ptychographic measurements. If we update the initial STEM probe to include these experiments we get

$$\Psi_0(\mathbf{k}) = A(\mathbf{k}) \exp[i\Phi(\mathbf{k})], \quad (17.7)$$

where $A(\mathbf{k})$ and $\Phi(\mathbf{k})$ are real-valued functions defining the amplitude and phase modulation respectively. You can easily construct an amplitude plate by taking a material which is opaque to fast electrons (typically a thick metal film) and milling apertures (holes) into it wherever you want electrons to pass through.

So, in summary, we remind you again that our electron probes are formed by aperture plates, typically containing a single circular hole. We place these apertures in one or more condenser lens planes, where one of them will act as the final probe-forming aperture which defines the maximum α_{probe} . For an ideal probe, a larger value of α_{probe} will always lead to a sharper probe, enabling higher-resolution imaging. However, as we just discussed, we use a probe-forming aperture because eventually the probe size will be limited by aberrations rather than by the diffraction limit. At 300 kV, these limits are roughly $\alpha_{\text{probe}} = 10$ mrad for an uncorrected microscope, 30 mrad for a C_3 -corrected instrument, and 50 mrad for a C_5 -corrected instrument. You can see examples of various probe-forming apertures in  Fig. 17.2.

We can also create more complex amplitude plates, for example adding patterns such as bullseyes to our STEM probe. This allows us to determine the position of each diffracted Bragg disk with higher precision and can also aid with disk position and intensity measurements from thick specimens, where dynamical diffraction can add artifacts to the DPs. Another

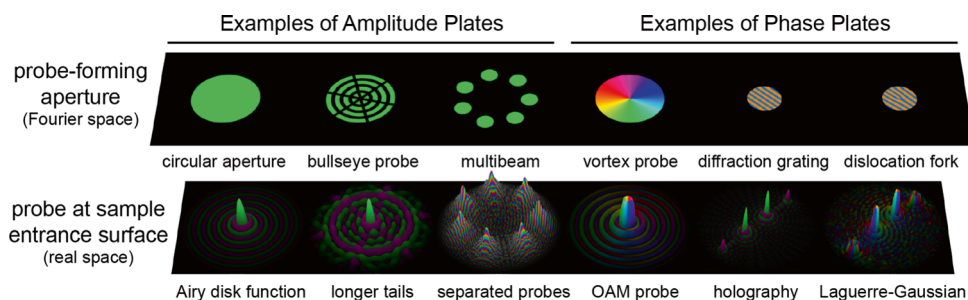


Fig. 17.2 Various STEM probes which can be formed using apertures with amplitude and phase plates

use of amplitude plates is in multibeam electron diffraction (MBED) where multiple STEM probes are formed, separated in momentum space. MBED can probe a wider range of scattering angles than conventional NBED and gives us some limited 3D information about our specimen because the different beams are incident on the specimen at different angles.

Phase plates in STEM are somewhat more complex than amplitude plates. (In previous chapters we've discussed how phase plates are common in visible-light microscopy but are only recently being developed for TEMs). 4D-STEM has driven the development of electron phase plates, which now fall into three broad classes: those composed of physical media, those formed from electric or magnetic fields, and those created using intense light fields. We can more easily machine physical phase plates to any desired geometry, but they will always cause amplitude changes in the probe, as well as unwanted scattering. We can use these types of phase plates for a variety of applications including aberration correction, adding phase diversity for ptychographic reconstruction, creation of complex wavefunctions such as vortex or Hilbert beams, and STEM holography where the probe is split into multiple beams using diffraction. They can also enable new contrast modes, for example enhancing interference between diffracted and un-diffracted (direct) beams in matched illumination and detector interferometry (MIDI) or annular differential phase contrast (ADPC) STEM. These kinds of phase plates can also be created in a more complex manner, for example by using a fast laser to write a pattern in charge on a surface which the beam passes through.

Alternatively, if we modulate the phase of the probe wavefunction using fields we will prevent unwanted scattering and amplitude modulation of the probe. However, we find that making small features or sharp phase discontinuities using fields is practically impossible, thus limiting the complexity of the phase-plate modulation. These fields can be generated using one, two, or even higher numbers of terminals to gener-

ate electric fields across the electron path, or by using electromagnetic fields such as those generated by a magnetic needle. In practice we combine both electric and magnetic field phase plates with amplitude elements, at a minimum usually a circular aperture to limit the maximum scattering angle.

17.3.5 Probe Step Size and Scan Patterns

One of the most powerful aspects of 4D-STEM experiments is that we can use very converged (small diameter and large α_{probe}) probes to measure properties on the atomic scale but take arbitrarily large or small steps between adjacent STEM probes. This means that when we are measuring material properties such as degree of crystalline order, phase, orientation, or local deformation, we can select a field of view which matches the functional length scale of the material or device being imaged. We could also carefully measure specimen damage from the beam, and then use a probe step size large enough to either not cause beam damage, or large enough to move into undamaged regions for each subsequent measurement. With more advanced scan-coil controllers, we can also change the scan pattern, skip some measurements for sparse sampling, or even use adaptive scan patterns.

For HR 4D-STEM experiments such as atomic-resolution imaging, we must be careful to sample well below the Nyquist sampling limit of the specimen structure. This also necessitates using a sufficiently converged (small) STEM probe such that we are below the Rayleigh criterion to resolve the features of interest. The sampling requirements can be equally stringent for computational-imaging methods such as ptychography, as these require sufficient overlap of adjacent probes to phase the object wavefunction we are reconstructing, and we discuss these in more detail in §17.3.7 and 17.6.3.

We must also take care to not over-sample our measurements. In addition to the risk that too much electron

fluence could potentially cause specimen damage, it may generate unwanted motion of the STEM probe with respect to the specimen. This movement can be caused by thermal motion of the specimen, charging of the specimen or microscope, or electronic instabilities of the instrument, all of which can lead to drift artifacts in the measurements. Specimen drift particularly affects HR 4D-STEM measurements, as the detector is often operated at a much slower speed than typical single-pixel ADF or BF detectors. The simplest method to overcome STEM drift is to record multiple overlapping datasets. We can then measure the distortions between these datasets, apply corrections, and correct these artifacts. This approach works even better when the STEM scan direction is rotated between adjacent measurements (i.e. we change the fast-scan direction of the STEM coils). Additionally, because we often record an ADF signal simultaneously with a 4D-STEM dataset, we could record additional fast-ADF scans and use these to calibrate and correct the 4D-STEM dataset.

17.3.6 4D-STEM as a Universal Detector

If we select a camera length such that we record a large fraction of the diffracted STEM probe from a 4D-STEM experiment, we can then create virtual detectors to form any imaging configuration we wish. We perform this operation by generating a numerical detector function, and then looping through all probe positions $\mathbf{R} = (x, y)$ and evaluating our detector function. This reduces our 4D dataset to one or more 2D-output images. These configurations include simple annular or circular detectors (analogous to traditional ADF and BF), detectors matched to the specimen signal such as integrating a subset of Bragg diffraction peaks, or even more complex detector shapes. We are also not limited to simple summation of a subset of pixels; we can multiply each pixel by a real value (or even a complex value as we wish) before the summation. Generating such an image $M(\mathbf{R})$ can be summarized as

$$M(\mathbf{R}) = \int_{\mathbf{k}} I(\mathbf{k}, \mathbf{R}) D(\mathbf{k}) d\mathbf{k}, \quad (17.8)$$

where $I(\mathbf{k}, \mathbf{R})$ is the 4D-STEM dataset covering diffraction-space coordinates $\mathbf{k}$ and real-space probe positions $\mathbf{R}$, and $D(\mathbf{k})$ is a detector function with one (potentially complex) value for each diffraction-space pixel. We can also calculate highly efficient detectors by using PCA or the related non-negative matrix factorization (NNMF). By using PCA or NNMF, we can calculate detectors which maximize contrast from different specimen regions, and de-noise the resulting images. We find that for large 4D-STEM datasets, these decompositions may not be feasible to calculate due to memory and

time constraints. To make the calculations more tractable, we could first bin the datasets, or radially integrate them in diffraction space to reduce the dimensions from four to three.

17.3.7 Electron Fluence and Beam Damage

As you well know, the topics of specimen damage due to the beam and optimizing the electron fluence used are very large subjects on their own. Here we will only discuss a few considerations specific to 4D-STEM and you can take a look back at ► Chap. 5 in this volume for more information. Virtually all our TEM specimens will be changed or damaged by the beam to some degree and this is the ultimate limitation to every TEM experiment. To perform a successful 4D-STEM experiment, we should take care to estimate the ‘critical dose’ where our specimen is damaged beyond our ability to make a reliable measurement, and then perform all experiments well below this dose level. You should note that the electron fluence is often colloquially referred to as the ‘electron dose’ in TEM experiments, but the correct term is fluence as we discussed in §5.2.1. Dose refers to the energy deposited per unit area, while fluence refers to the number of measurement particles per unit area.

In many 4D-STEM experiments with modern detectors, we can count all the electrons at each probe position. This measurement gives us the mean electron density per electron probe D_p . If our measurement has a step size between adjacent probes p , then the fluence (electrons per unit area) is equal to $D_c = D_p/p^2$. Note that this is the mean number of electrons per unit area; if we have an overly focused electron probe, the ratio R of the maximum number of electrons per unit area to the mean is approximately equal to

$$R = 1 + \pi p^2 k_{\text{probe}}^2 \exp\left(-k_{\text{probe}}^4 \lambda^2 |C_1|^2\right), \quad (17.9)$$

where C_1 is the defocus (C_1 is typically equal to -1 times the microscope defocus Δf). In practice, you’ll find that this expression is not very useful for estimating the degree of overlap between adjacent probes. Instead, we can consider our probe as a perfect cone emanating from the crossover with a semi-angle α_{probe} . In the specimen plane at a distance $|C_1|$ from the probe crossover, the probe shape is approximately given by a circle of diameter $2 \alpha_{\text{probe}} |C_1|$. Given this simplification, the fractional overlap between adjacent probes is approximately

$$O_{\text{probe}} = 1 - \frac{p}{2 \alpha_{\text{probe}} |C_1|}. \quad (17.10)$$

Therefore, we can use the mean electron-fluence approximation as long as $p \ll 2\alpha_{\text{probe}}|C_1|$. You'll see that precession is also the imaging condition required for defocused ptychographic imaging, which we'll describe in more detail below.

17.3.8 Precession Electron Diffraction (PED)

One of the common limitations in 4D-STEM studies of crystalline materials is that thicker specimens will scatter the beam multiple times. This leads to strong intensity modulations which complicate the analysis. One of the most common ways to circumvent these issues is to use PED which we introduced back in ► Chap. 8. In a PED experiment, we apply a tilt to the beam, continuously rotating the tilt vector around the optic axis during the 4D-STEM scan. As long as the rotation frequency is faster than the detector frame rate, PED experiments will average the diffraction contrast from a cone of projection angles, with semi-angles typically in the range $0.5^\circ < \alpha_{\text{tilt}} < 3^\circ$. The ideal PED diffraction intensity $I_{\text{PED}}(\mathbf{k})$ can be modeled as

$$I_{\text{PED}}(\mathbf{k}) = \frac{1}{2\pi} \int_0^{2\pi} I(k_x - k_{\text{tilt}} \cos \phi, k_y - k_{\text{tilt}} \sin \phi) d\phi, \quad (17.11)$$

where $I(\mathbf{k})$ is the DP measured using the incident scattering vector $\mathbf{k}$, the precession scattering vector is equal to $k_{\text{tilt}} = \alpha_{\text{tilt}}/\lambda$, and ϕ is the in-plane rotation angle of the beam. If you're doing PED experiments, you'll need to carry out a very careful alignment of your TEM; you'll need to align both the cross-over in the specimen plane to keep the beam as focused as possible and apply the correct amount of de-scan in diffraction space below the specimen so that the DPs do not shift as a function of ϕ . You can also perform serial PED

experiments, where you sequentially record a series of images for different beam tilts. This procedure requires significantly more data processing, but you can correct both real space and diffraction space shifts during post-processing. Additionally, if you wish, you can replace the above averaging operator with a more robust median operator which is better at rejecting diffraction-intensity outliers.

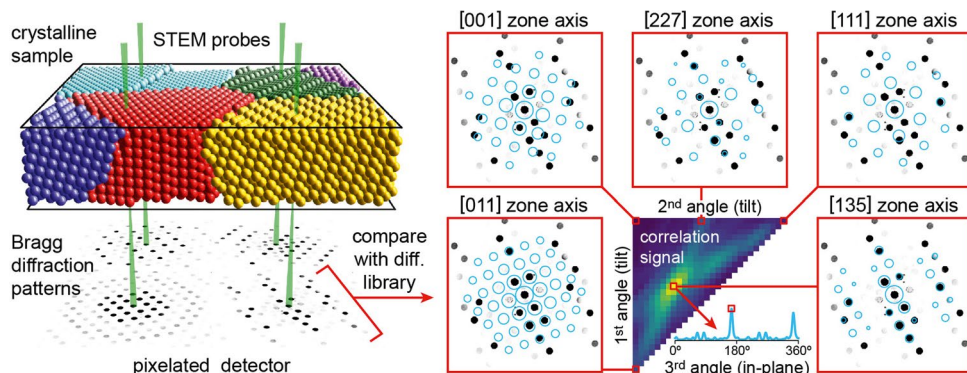
As you know, PED reduces the dynamical effects from multiple scattering in most cases, but typically cannot completely remove them. One alternative to PED that we discussed above is to use patterned amplitude plates in the probe-forming aperture or to use MBED to perform 'precession without precessing' experiments. PED tends to perform better for orientation-mapping experiments, while patterned apertures have a slight edge for strain mapping. MBED requires significantly more data processing than either PED or patterned apertures, both of which we can typically analyze using the same software as conventional NBED experiments.

17.4 NBED of Crystalline Specimens

17.4.1 Orientation Mapping of Crystalline Materials

Suggestion

If you've been doing TEM diffraction for a while and successfully negotiated ► Chaps. 7, 8, 9 in this book, then you'll find this section is really basic and covered back in W&C ► Chaps. 11, 12 and 13 so you can jump ahead to see how ► Fig. 17.3 makes classical DPs much easier to use.



► Fig. 17.3 Automated crystal orientation mapping (ACOM) workflow. After computing a library of symmetry-unique DPs, we compute the correlation of each experimental diffraction image as a function of the 3 Euler angles, angles 1 and 2 corresponding to the zone axis/specimen tilt, and angle 3 corresponding to in-plane rotation. We assign each probe position to the orientation with the highest correlation signal. Note that we must also consider in-plane mirror operations

Here's a bit of revision before we leap into the good stuff. Common specimens which we can study with NBED experiments are single crystal or polycrystalline. As you know, Bragg diffraction from a crystalline lattice produces very sharp characteristic diffraction maxima which typically have high intensities (lots about this in W&C ► Chap. 13—but if you've read this far then you probably don't need to check it out). The lattice measured in diffraction space acts as a fingerprint for the underlying crystal structure and orientation. To perform orientation mapping in 4D-STEM, we first calculate a 'library' of DPs from a crystal structure at each symmetry-unique orientation. We next compute a correlation function between our experimental DPs and our library, and finally assign one or more matches with the highest correlation values. Again, from Crystallography 101, you know that a crystal is defined primarily by a unit cell, given by its three edge vectors ($\mathbf{a}, \mathbf{b}, \mathbf{c}$). To form a crystalline lattice, this unit cell is tiled infinitely in all directions. To calculate a DP (see ► Chap. 12 in W&C) we first calculate the reciprocal-lattice vectors ($\mathbf{a}, \mathbf{b}, \mathbf{c}$) given by the reciprocal of the real-space lattice, equal to

$$\begin{aligned} \mathbf{a} &= \frac{\mathbf{b} \times \mathbf{c}}{\Omega} \\ \mathbf{b} &= \frac{\mathbf{c} \times \mathbf{a}}{\Omega} \\ \mathbf{c} &= \frac{\mathbf{a} \times \mathbf{b}}{\Omega} \end{aligned} \quad (17.12)$$

where $\Omega = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$ is the unit cell volume, and $\times$ is the vector cross product operator. Each reciprocal lattice point is a possible location of a Bragg diffraction spot, given by

$$\mathbf{g}_{hkl} = h\mathbf{a} + k\mathbf{b} + l\mathbf{c} \quad (17.13)$$

where (h, k, l) are integer Miller indices. Each one of these lattice points has a complex structure factor, which taken together fully describes the electrostatic potential inside the unit cell. The number of electrons which will be scattered into a given Bragg vector at one of these lattice planes is proportional to the structure-factor intensity (amplitude squared) of each of these points. Given the species and fraction positions $\mathbf{p}_n$ of N atoms inside the unit cell, we compute the structure factors F_{hkl} as

$$F_{hkl} = \frac{1}{\Omega} \sum_{n=1}^N f_n(|\mathbf{g}_{hkl}|) \exp[-2\pi i \mathbf{p}_n \cdot (h, k, l)] \quad (17.14)$$

where $f_n(k)$ are the atomic-scattering factors for atom n , which are tabulated values for each atomic species. The mathematical construction we use to calculate a DP is the Ewald sphere which is discussed in more detail in ► Chap. 12 of W&C. If we tilt a sphere with radius $1/\lambda$ to a diffraction vector direction $\mathbf{k}$, then each point on or very close to the sphere surface will appear in the DP as a bright spot. If the specimen surface is

tilted an angle α away from the beam direction, the distances of each reciprocal lattice point from the sphere is called the "excitation error" s_g and is given by

$$s_g = \frac{-\mathbf{g} \cdot (2\mathbf{k} + \mathbf{g})}{2|\mathbf{k} + \mathbf{g}| \cos \alpha} \quad (17.15)$$

The reason the lattice points are not infinitely sharp is because they are convolved the specimen shape factor, which is the FT of the specimen geometry. Because real specimens must be thin enough to allow the beam to pass through them, the thickness t between the top and bottom surfaces creates a so-called *relrod* shape which we can use to calculate the intensity of each diffraction spot from a thin specimen as

$$I_{hkl} = |F_{hkl}|^2 \frac{\sin(\pi s_g t)}{\pi s_g} \quad (17.16)$$

and find the position of each diffraction spot using the geometrical projection onto the Ewald sphere surface. The procedure for performing 4D-STEM orientation mapping using this approach is shown in ► Fig. 17.3.

One limitation of 4D-STEM orientation mapping is that because our wavelength is high, the Ewald sphere is very flat. This means that we can only sample a small angular range of reciprocal space, leading to possible ambiguities in our orientation maps. Generally, 4D-STEM-orientation mapping has lower angular accuracy than EBSD in SEM, but a higher spatial resolution. Another limitation you'll find is that it may be difficult to distinguish orientations with similar DPs, for example unit cells which have a lattice-vector length close to an integer multiple of their other lattice-vector lengths. These small differences in scattering-vector length may be below the calibration limitations of either the absolute values or pattern distortions such as ellipticity from an imperfectly aligned projector system in the STEM. So here's yet another good reason to make sure your (S)TEM is very well aligned before doing any 4D STEM.

Another potential limitation of orientation mapping is that if your specimen is thick or contains heavy elements it will scatter the beam multiple times. This creates oscillating intensities in your DPs, making the matching procedure more challenging. Such a dynamical-diffraction effect is especially pronounced if your crystal is tilted to a low-index zone axis, or when a larger α_{probe} is used because in this case there may be significant intensity variations inside each diffracted disk. As you now know, the best way to reduce this artifact is to use PED. Because the diffraction intensities are averaged over multiple orientations, they approximate a more kinematical (single scattering) set of intensities. PED also somewhat reduces the limitations

of the flat Ewald sphere that we discussed above, as it probes a range of incident directions on the specimen as opposed to a single direction. Using PED will however reduce the spatial resolution of our orientation maps, and so we must again balance the accuracy and precision of our reciprocal-space measurements against the real-space resolution.

17.4.2 Phase Mapping of Crystalline Materials

Phase Mapping vs. Phase of Wavefunction

Note that here in §17.4.2, we use ‘phase’ to refer to a specific structural and chemical phase. The phase of a complex-valued electron wavefunction as described in §17.3.2 and others is a separate topic, similar only in name.

One of the most straightforward extensions to NBED orientation mapping is crystalline-phase mapping. First, we compute a diffraction library for each potential crystal structure, and then perform the above orientation mapping for each structure. We can then compare the correlation matching signals between different phases, and assign the phase detected (if any) for each probe position. You’ll find that this procedure is straightforward when the two crystals have very different unit-cell geometries, but it may be quite challenging to separate structures which produce similar DPs. This challenge is analogous to the orientation ambiguity between similar directions that we discussed above. This issue can be even more difficult to overcome when the two crystal structures share lattice vectors, as is often the case for epitaxial film growth or coherent precipitation in alloy systems.

A promising extension to phase mapping from diffraction data is to incorporate simultaneous composition measurements using XEDS. Combining elastic diffraction with local composition often makes it easier to distinguish multiple phases. In principle these measurements could be performed sequentially, but aligning dissimilar data channels is difficult in practice.

17.4.3 Strain Mapping of Crystalline Materials

We can also perform precision measurements of the local deviations of the lattice constants of crystalline materials using 4D-STEM. These measurements are usually given in terms of the strain tensor, a 2D matrix

which describes the local deformation of the crystalline lattice with respect to a reference lattice. This reference can be provided externally from a known crystal structure, measured from another 4D-STEM dataset, or even taken from a subset of the same 4D-STEM scan. We can use this last method for example to measure the strain of a thin film with respect to the substrate on which it has been grown epitaxially.

As you know, Bragg-diffraction experiments of a crystal tilted along a low-index zone axis produce a regular 2D grid of diffraction spots. In a typical 4D-STEM strain measurement, we first measure the position of all diffraction spots with as high precision and accuracy as possible. (You can enhance the precision by using patterned apertures or using PED, though you should keep in mind that both additions will also reduce the spatial resolution of your measurement.) Next, we characterize the positions $\mathbf{g}_{ab} = (g_x, g_y)$ of our 2D grid of spots using the lattice equation.

$$\mathbf{g}_{ab} = \mathbf{g}_0 + a\mathbf{u} + b\mathbf{v}, \quad (17.17)$$

where $\mathbf{g}_0$ is the origin of our lattice (usually positioned on the direct beam), $\mathbf{u}$ and $\mathbf{v}$ are non-colinear lattice vectors defining the grid, and (a, b) are rational indices. Ideally (a, b) will be integers or fractions with a small denominator, so that we can unambiguously index all the diffraction spots and exclude any false positives. Typically, we have an initial guess of the six lattice coefficients $(\mathbf{g}_0, \mathbf{u}, \mathbf{v})$, which we can compare to the measured spot positions $\mathbf{g}_{ab}$ to estimate the values of (a, b) for all spots using linear algebra. Next, we calculate the best-fit lattice coefficients, also using linear algebra. This fitting can be weighted by the estimated diffraction-spot intensities to improve the accuracy, or we can employ an iterative, robust, least-squares procedure to exclude the influence of outliers. If we repeat this fitting for all probe positions $\mathbf{R}$, we get sets of lattice vectors $\mathbf{u}(\mathbf{R})$ and $\mathbf{v}(\mathbf{R})$. The origin fits can be used as a rigid-shift input for DPC measurements which we’ll describe in §17.6.1 but these are not used for strain measurements.

Next, we define our reference lattice with the vectors $\mathbf{u}_0$ and $\mathbf{v}_0$. For each probe position $\mathbf{R}$, we calculate a transformation matrix which maps the reference lattice onto the measured lattice. This transformation is defined as

$$\begin{bmatrix} a(\mathbf{R}) & b(\mathbf{R}) \\ c(\mathbf{R}) & d(\mathbf{R}) \end{bmatrix} \begin{bmatrix} \mathbf{u}_0 \\ \mathbf{v}_0 \end{bmatrix} = \begin{bmatrix} \mathbf{u}(\mathbf{R}) \\ \mathbf{v}(\mathbf{R}) \end{bmatrix}, \quad (17.18)$$

where (a, b, c, d) are the 4 elements of the transformation matrix. We then solve this equation for equal probe position using linear algebra. If the measured lattice is unstrained, we will recover the identity ma-

trix ($a = d = 1, b = c = 0$). Note that we are working in reciprocal space, which means that, in the infinitesimal strain limit, the real-space strains will be equal to the negative value of the strains measured in reciprocal space. With this in mind, we can define strains maps as

$$\begin{aligned}\varepsilon_{xx}(\mathbf{R}) &= 1 - a(\mathbf{R}) \\ \varepsilon_{yy}(\mathbf{R}) &= 1 - d(\mathbf{R}) \\ \varepsilon_{xy}(\mathbf{R}) &= -\frac{1}{2}c(\mathbf{R}) - \frac{1}{2}d(\mathbf{R}) \\ \phi(\mathbf{R}) &= \frac{1}{2}c(\mathbf{R}) - \frac{1}{2}d(\mathbf{R})\end{aligned}\quad (17.19)$$

where $\varepsilon_{xx}(\mathbf{R})$ and $\varepsilon_{yy}(\mathbf{R})$ are the infinitesimal strains in the x and y directions respectively, $\varepsilon_{xy}(\mathbf{R})$ is the shear strain, and $\phi(\mathbf{R})$ is the infinitesimal rotation. You'll find that strain mapping in 4D-STEM is more resilient to calibration errors such as ellipticity than other NBED property measurements. This is because we are often measuring strains with respect to a measured reference lattice, and so both lattices will often contain the same distortions. This is especially true of the reference lattice is taken from a subset of the scan in question.

Note, however, that we must take care that our real-space STEM scanning directions are correctly rotated with respect to our diffraction-space coordinate system, otherwise our strains may be rotated to a new coordinate system. We can correct this artifact by rotating the initial Bragg peaks, or by applying a tensor rotation to the strain outputs. If the rotation value $\phi(\mathbf{R})$ is larger than the small angle limit, we should replace the above infinitesimal strain tensor with a finite strain tensor. The finite strain tensor can be calculated using a polar decomposition of the transformation matrix and we show the full workflow of a strain mapping experiment in Fig. 17.4.

4D-STEM strain mapping can be combined with the orientation mapping as you just read in §17.4.1. In this case, we use our best-fit orientation as the refer-

ence lattice and calculate the deformation matrix with respect to this lattice. In this way we can avoid needing global reference vectors applied to the full field of view, which may be especially helpful when our specimen is composed of polycrystalline grains with unique orientations. Strain mapping can also be extended to 3D by combining deformation-matrix measurements from multiple projection directions. This allows us to estimate the full six components of the 3D strain tensor, rather than the only three components which we can estimate from a 2D projection measurement.

There are also alternative methods of estimating strain from DPs, including using a cepstral transform. For each Bragg DP $I(\mathbf{k}, \mathbf{R})$ at probe position $\mathbf{R}$ with high SNR, we can compute the cepstral transformation $C(\mathbf{r}, \mathbf{R})$ as

$$C(\mathbf{r}, \mathbf{R}) = |\mathcal{F}_{\mathbf{k} \rightarrow \mathbf{r}}\{\log(I(\mathbf{k}, \mathbf{R}) + I_0)\}|, \quad (17.20)$$

where I_0 is an intensity offset to prevent us having to take the logarithm of zero. We need the logarithmic transformation for thin specimens to scale the Bragg peaks to have similar intensities (especially compared to the 000 beam), though it can be omitted for thicker specimens where all lower-order peaks may have similar intensities. From the resulting spectrum image $C(\mathbf{r}, \mathbf{R})$, we then choose two peaks and use subpixel fits of their position to estimate the real-space lattice vectors. We can then compute strain maps from these lattice vectors using the same reference lattice procedure we described above. This method of 4D-STEM strain mapping is similar to using the geometric phase analysis (GPA) method to calculate strain as described in ► Chap. 9.

We can make 4D-STEM strain mapping significantly more accurate and precise if we use a probe formed by a very small α_{probe} or patterned-amplitude apertures. Both methods, however, will increase the probe size in real space, and therefore reduce our spatial resolution. There is no free lunch—inevitably *you* must

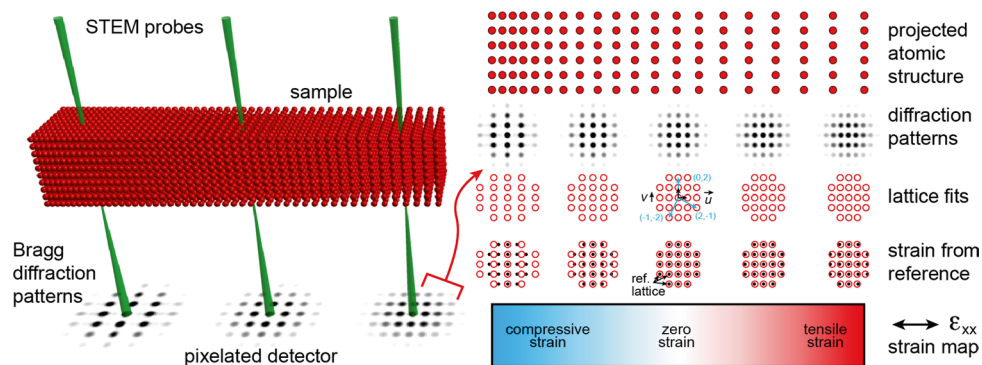


Fig. 17.4 4D-STEM strain mapping workflow. For each DP, we fit a lattice to all of the diffracted Bragg peaks. This first requires each peak be assigned to a lattice vector, and then calculating the best fit lattice vectors. Finally, we choose reference strain vectors (often from a reference region of the specimen), and then calculate the strain transformation between the reference and measured lattices for each pattern

decide how to resolve the tradeoff between real-space resolution and diffraction-space accuracy depending on what you want to find out about your specimen. A similar compromise is needed if you're using PED in strain mapping. PED may improve the accuracy by decreasing artifacts due to strong dynamical diffraction but will also reduce the precision of 4D-STEM measurements performed in diffraction space as the Bragg signals will become more blurry.

17.5 NBED of Disordered Specimens

Many interesting materials are disordered, ranging from nanocrystalline materials to amorphous to liquids. Diffraction from these materials usually produces patterns known as 'amorphous halos', which are approximately circular DPs resulting from the lack of translational symmetry of the specimen. The local atomic structure of disordered materials is usually characterized by the degree of ordering, and the length scale(s) over which this ordering persists. We can probe the local structure of these materials using NBED in a variety of related ways. These methods tend to follow the same pattern; we record a series of DPs which contain amorphous halos, then we use computational analysis to estimate some portion of the material's structure factors, and then invert these estimates into the property of interest. Energy-filtered 4D-STEM, where we place the detector behind a spectrometer and use a slit to select the zero-loss beam, can be quite valuable because it removes all the background signals. The radially symmetric inelastic backgrounds due to plasmon scattering can often mimic or distort the structure factor signals so removing them eliminates this danger.

17.5.1 Pair Distribution Functions

Measuring the atomic pair distribution function (PDF) is a classic diffraction method to characterize disordered materials. By using 4D-STEM, we can measure variations in the PDF as a function of position in the specimen, for example approaching interfaces, surfaces, or over chemical gradients. In 4D-STEM PDF measurements, we use a near-parallel beam. Typical α_{probe} values used for these experiments are 0.1–0.2 mrad. We also need to ensure that we use a short enough camera length that we capture the high-angle scattering. It is very important to measure the DPs out to high enough angles that we capture the asymptotic single-scattering, atomic form factors, beyond the oscillations of the structure factor. To perform quantitative PDF mapping, we typically require a thin enough specimen that most electrons are scattered only once (the kinematical-scattering regime), and energy-filtered dif-

fraction to remove unwanted background signals such as inelastic scattering from bulk and surface plasmons.

The mathematical basis of the PDF method follows the work of Debye. The scattering amplitude $A(\mathbf{k})$ from atom j is equal to $A(\mathbf{k}) = f_j(k) \exp(i\mathbf{k} \cdot \mathbf{r}_j)$, where $f_j(k)$ is the atomic scattering factor. Applying the Debye formula to an amorphous material composed of N atoms of the same species, and then integrating over the full specimen volume V gives us a radially symmetric diffraction intensity $I(k)$ of approximately.

$$I(k) = Nf^2(k) + f^2(k) \sum_j \int 4\pi r_j^2 [g(r_j) - \rho_0] \frac{\sin(qr_j)}{qr_j} dV, \quad (17.21)$$

where $g(r_j)$ is the PDF for atomic bonds from the origin to an atom j at radius r_j , and ρ_0 is the mean density. The first step to extract $g(r_j)$ from the radially averaged DP intensity $I(k)$ is to estimate the reduced structure factor.

$$S(k) = \frac{I(k) - Nf^2(k) - B(k)}{Nf^2(k)}, \quad (17.22)$$

where $B(k)$ is the background intensity. In an energy-filtered 4D-STEM experiment where we've calibrated the detector with a good dark reference, then ideally $B(k) = 0$. We can use look-up tables to estimate the $Nf^2(k)$ term from simulated-scattering factors but, in practice, fitting a polynomial, power law, or some other function to the form factors works better for real experiments (which often contain multiple atomic species). We suggest estimating the reduced structure factor using the function.

$$S(k) = \frac{I(k) - A_2 \exp(-k^4/\sigma_2^4) - A_1 \exp(-k^2/\sigma_1^2) - A_0}{A_2 \exp(-k^4/\sigma_2^4) - A_1 \exp(-k^2/\sigma_1^2)} k, \quad (17.23)$$

where $(A_2, A_1, A_0, \sigma_2, \sigma_1)$ are fitting coefficients. Once we have this estimate, we can recover the reduced density function by computing.

$$G(r) = \frac{4}{\pi^2} \int_0^\infty S(k) \sin(kr) dk. \quad (17.24)$$

Note that you should numerically evaluate this integral, so that you can estimate the PDF using your desired range and sampling for r . When evaluating this integral, take care to include the sampling Δk of the k dimension! The reduced density function (not to be confused with the radial distribution function!) obeys the relationship $G(r) = 4\pi r \rho_0 [g(r) - 1]$, and therefore we can numerically calculate the PDF using the expression.

$$g(r) = \frac{G(r)}{4\pi \rho_0 r \Delta r} + 1, \quad (17.25)$$

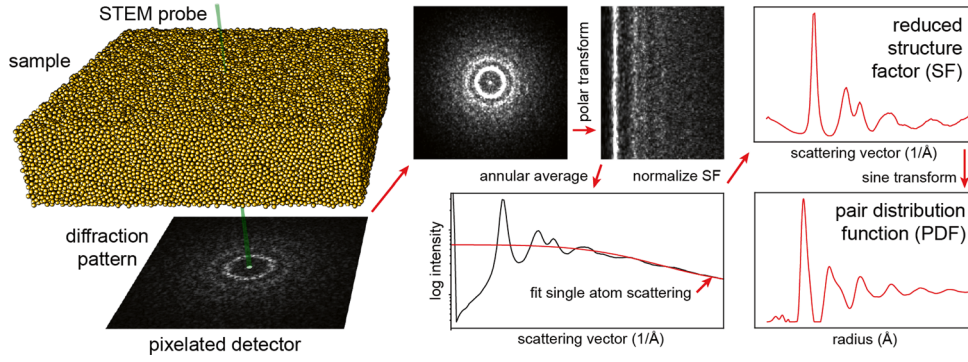


Fig. 17.5 4D-STEM PDF mapping workflow. First, we record DPs with a small α_{probe} out to large scattering angles. Next, we calibrate the center coordinate and ellipticity of each pattern, and then calculate the polar transform and average along the annular direction. We then fit the single atom scattering and use it to calculate the reduced structure factor. Finally, we apply a sine transformation and normalization to reconstruct the PDF

where Δr is the sampling along the r dimension. In most experimental PDF estimates, we will see substantial errors as $r \rightarrow 0$. We can either ignore these errors or remove them in one of two ways. The first method is to find r corresponding to the first maxima in $g(r)$, and multiply a damping function which goes to 0 as $r \rightarrow 0$. The second method is to carefully high-pass filter $S(k)$, though this can be difficult in practice. You can see the PDF workflow schematically in [Fig. 17.5](#).

The primary limitation in 4D-STEM measurements which spatially map the PDF is that to increase the spatial resolution, we need to use a larger α_{probe} , leading to a smaller probe size. This however will decrease the accuracy of our PDF measurement. You must carefully choose the probe size by considering the trade-off between real and reciprocal -space accuracy. One advantage of 4D-STEM is that you'll have a large number of PDF measurements, which allows you to evaluate the changes in the PDF distributions as a function of probe position. The other challenges that you'll face include the previously mentioned unwanted background signals due to inelastic scattering, and the breakdown of equation Eq. 17.21 due to multiple scattering in thicker specimens.

17.5.2 Fluctuation Electron Microscopy

The PDF discussed above is often described as measuring short-range order (SRO). However, many disordered materials such as metallic glasses have strong ordering beyond the first few neighboring shells of atoms. This property is usually called medium-range order (MRO), in contrast with the SRO of highly amorphous materials or liquids and the long-range order (LRO) of crystalline specimens. To probe the MRO of a specimen, we expand the probe α_{probe} , reducing the probe size until it matches the ordering cluster size

of the material. This leads to diffraction signals which are often referred to as 'speckle patterns', where we see strong oscillations along the annular direction in the amorphous halos. These 'fluctuations' give this method its name, fluctuation electron microscopy (FEM). To evaluate the FEM signal, we transform the cartesian DPs $I(k_x, k_y)$ to polar coordinates $I(k, \phi)$, and then computing the annular average intensity $I(k)$. We then measure the mean intensity $M(k)$ and variance $V(k)$ over the probe positions $\mathbf{R}$. Finally, we average these measurements over all probe positions, giving the expressions.

$$M(k) = I(k)_{\mathbf{R}}, \quad (17.26)$$

$$V(k) = [I(k) - I(k)_{\mathbf{R}}]^2_{\mathbf{R}}, \quad (17.27)$$

where $_{\mathbf{x}}$ represents taking the average along variable x . Note that we can also define a 'local' FEM measurement, where we instead compute the variance inside each radial bin. We can then either average the resulting measurement over the probe positions $\mathbf{R}$, or map the FEM signal as a position of $\mathbf{R}$ giving

$$M(k, \mathbf{R}) = I(k, \phi, \mathbf{R})_{\phi}, \quad (17.28)$$

$$V(k, \mathbf{R}) = [(k, \phi, \mathbf{R}) - I(k, \phi, \mathbf{R})_{\phi}]^2_{\phi}. \quad (17.29)$$

For both types of FEM measurements, the final output is usually the normalized variance given by

$$V_{\text{norm}}(k) = \frac{V(k)}{M(k)^2}. \quad (17.30)$$

While the PDF function measures two-body pair correlations, the FEM signal is more sensitive to four-body, pair-pair correlations. There are diffraction studies demonstrating that similar, yet distinct, disordered materials can produce near-identical PDF signals but you can tell them apart using FEM. Additionally, we can

measure the FEM signal as a function of α_{probe} ; this allows us to plot the variance as a function of probe size, implicitly related to the cluster size of the material and therefore to the length scale of the MRO.

Like PDF measurements in 4D-STEM, FEM experiments are often more reliable when you are measuring difference signals between similar specimens. This could apply to specimens with very similar compositions, or measurements from the same material with different processing histories, or *in-situ* experiments where you measure the FEM signal over time.

17.5.3 Angular Correlation Functions

It gets even more complicated when we study local order in amorphous materials. Such order can be described by correlation functions which are more complex than simple PDFs or the normalized variance of FEM. The most complex of these are n-body distribution functions which can be evaluated with transformations based on spherical harmonics. A more straightforward method, which we will describe here, is evaluation of angular correlations from DPs. Specifically, we mean measuring the degree of n-fold rotation signal from a ring at a given scattering angle, to infer information about the local atomic symmetries within the specimen. For example, an octahedral cluster of atoms imaged along a vertex, edge, or face will produce diffraction intensities with stronger fourfold, twofold and sixfold correlations respectively.

We must first transform the diffracted intensity in cartesian coordinates $I(k_x, k_y)$ to polar coordinates $I(k, \phi)$. Then we can evaluate a normalized angular cross-correlation function using the expression.

$$C(k, \mathbf{R}, \Delta\phi) = \frac{I(k, \phi, \mathbf{R})I(k, \phi + \Delta\phi, \mathbf{R})}{I(k, \phi, \mathbf{R})^2} - 1, \quad (17.31)$$

where $I(k, \phi + \Delta\phi, \mathbf{R})$ is a DP at position $\mathbf{R}$ in polar coordinates, shifted in the annular direction by an angle $\Delta\phi$. The denominator removes the dependence of the correlation signal on the total number of electrons scattered into a ring of radius k . We can evaluate Eq. 17.31 efficiently for all $\Delta\phi$ by using a FFT along the annular direction ϕ . Different structural units in amorphous materials such as tetrahedra, octahedra, icosahedra and others have unique angular distributions of correlations and can therefore be measured and differentiated in a statistical manner using angular correlations.

You may also find it helpful to interpret the angular correlation function by measuring the degree of sym-

metry for n-fold rotations, corresponding to angular correlations over angles of $360^\circ/n$. This operation can also be efficiently performed by computing the FFT along the annular direction ϕ . We must be careful, however, not to over-interpret these patterns, especially since scattering from thin specimens will typically obey Friedel's law and produce twofold rotational symmetry. We will also typically select α_{probe} using guidelines similar to those in the previous FEM section. We need an α_{probe} large enough to produce a strong speckle pattern, but not so large that the DPs become completely scrambled. For most amorphous materials characterized using 300 kV electrons, this α_{probe} range will be approximately 1–4 mrad. We also need to use thin specimens to achieve quantitative results; in addition to potential artifacts from multiple scattering, this method will measure many spurious correlations because in thick specimens there are a large number of structural units projected along the beam direction in each DP.

17.5.4 Strain Mapping of Amorphous Materials

As we've discussed, already, diffraction from amorphous materials will often contain amorphous halos. These rings are directly formed from the structure factor of the specimen, and are therefore related to the mean atomic spacing. If we mechanically deform an amorphous specimen, we can measure the local deformation from high-precision fitting of the amorphous halo in each probe position. It's important to note that all 4D-STEM measurements contain some degree of ellipticity, due to optical distortions in the projector lenses. Therefore, we must either measure changes in the ellipticity of the amorphous halo relative to a reference region of the specimen or to a reference scan. We can define a polar elliptic-coordinate system that we can fit in a numerically stable manner using an implicit representation defined as

$$k_e^2 = A(k_x - k_x^0)^2 + B(k_x - k_x^0)(k_y - k_y^0) + C(k_y - k_y^0)^2, \quad (17.32)$$

where k_e is the radial distance to an ellipse from an origin at the center coordinate (k_x^0, k_y^0) , and (A, B, C) are the ellipse-fitting coefficients. You can see that we recover circular polar coordinates when $A = C = 1$ and $B = 0$. You'll find there are many different possible functions which can be used to fit an amorphous halo. One such function is to include a Gaussian-distribution background, and use a two-sided Gaussian distribution for the halo, giving

$$I(k) = I_0 \exp\left(\frac{-k_e^2}{2\sigma_0^2}\right) + I_{12} \left[\exp\left(\frac{(k_e - k_r)^2}{-2\sigma_1^2}\right) u(k_r - k_e) + \exp\left(\frac{(k_e - k_r)^2}{-2\sigma_2^2}\right) u(k_e - k_r) \right], \quad (17.33)$$

where I_0 and I_{12} are the background and amorphous-halo peak intensities respectively, k_r is the radial position of the amorphous halo, $(\sigma_0, \sigma_1, \sigma_2)$ are the standard deviations for the background, the inner portion, and the outer portion of the halo respectively, and $u(k_r - k_e)$ is the unit step function (or Heaviside step function) which is equal to 1 for positive values and 0 for negative values. To transform these fits into strain maps, we can linearly transform the polar-elliptic coefficients into strain coefficients, for all probe positions $\mathbf{R}$. If the strains are small (infinitesimal) we can use the expressions.

$$\begin{aligned} \varepsilon_{xx} &= [1 - A(\mathbf{R})]/2 \\ \varepsilon_{xy} &= -B(\mathbf{R})/2, \\ \varepsilon_{yy} &= [1 - C(\mathbf{R})]/2 \end{aligned} \quad (17.34)$$

where ε_{xx} and ε_{yy} are the strains along the x and y directions, and ε_{xy} is the shear strain. Slightly more complex equations can be used for larger strain magnitudes. Note that for the crystalline strain mapping described above, we can't estimate the local rotation of an amorphous specimen in 4D-STEM. This is because the DPs are approximately radially symmetric, and thus appear nearly identical for in-plane specimen rotations. We can, however, tilt our specimen relative to the optic axis, to probe the local atomic spacing in the out-of-plane direction. This technique has been used in experimental studies of thin amorphous films to characterize relative changes in the strain along the growth direction compared to the in-plane directions.

17.6 Phase Retrieval Imaging

Phase Retrieval vs Crystal Phase

Note that here in §17.6, we are reconstructing the 'phase' of a complex-valued electron wavefunction. This is a different topic to the crystalline phase mapping described in §17.4.2.

One of the most powerful uses for 4D-STEM is to enable high SNR and HR phase retrieval methods. We prefer to use these methods for imaging weakly scattering specimens such as biological materials, as it is much more dose-efficient than incoherent imaging methods

such as HAADF. HAADF is very popular in materials science due to its simple configuration and easy modeling and interpretation, but, as we've noted, it produces essentially no contrast for light elements including important species such as O, C and Li. Phase-retrieval methods in a Cs-corrected STEM however can image columns of light elements or, in some cases, single light atoms. Here we will cover the two more popular retrieval methods in 4D-STEM: DPC and ptychography. Various other methods to perform phase-retrieval STEM which we'll not discuss here include STEM holography, scattering matrix (or S-matrix) reconstruction, parallax /tilt-corrected BF, and Fourier ptychography.

17.6.1 Center-of-Mass Differential Phase Contrast

There are several ways our STEM probe can be deflected: electric fields due to impacting a gradient in our specimen's projected (electrostatic) potential, a gradient in an extrinsic electrostatic field, or by a gradient in an electromagnetic field. Of course, the latter two effects are why we can use electric and magnetic components to form lenses, deflectors, stigmators, and other optical elements in our TEMs and EELS. Here, we we'll focus just on using DPC to measure gradients in the projected potential of a specimen. When a beam of fast electrons passes through matter, it will be deflected by gradients in this potential, projected into the 2D plane normal to the beam. We can model an initial probe wavefunction $\psi_0(\mathbf{r})$ passing through our specimen potential consisting of an infinite plane wave with a wavevector equal to $\mathbf{k}_0$ by applying the equation

$$\psi(\mathbf{r}) = \psi_0(\mathbf{r}) \exp(-2\pi i \mathbf{k}_0 \cdot \mathbf{r}), \quad (17.35)$$

where $\mathbf{r} = (r_x, r_y)$ are the real-space coordinates, and $\psi(\mathbf{r})$ is the output wavefunction. The resulting deflection of the beam is more obvious in the far-field detector plane, which we can compute by taking the FT $\Psi(\mathbf{k}) = \mathcal{F}_{\mathbf{r} \rightarrow \mathbf{k}}\{\psi(\mathbf{r})\}$ of Eq. 17.35, giving

$$\Psi(\mathbf{k}) = \Psi_0(\mathbf{k} - \mathbf{k}_0). \quad (17.36)$$

Here you can see that a plane wave will cause a shift of the probe proportional to the wavevector $\mathbf{k}_0$, i.e. proportional to the specimen's electric field and equal to the gradient of the projected potential. You should always be aware, however, that in a real experiment, your STEM probe is not infinitely sharp. Because of non-zero size of the probe, real 4D-STEM DPC experiments will typically contain complex intensity redistributions in the DPs. Nevertheless, the simple physical picture that we've given above is enough to understand the basics of contrast formation in DPC.

As we just saw in ► Chap. 16, DPC has become a popular imaging method in STEM, because it can produce a phase-contrast image using a very simple detector configuration, typically a four-quadrant segmented detector. If we position a segmented detector at the outer edge of the central disk, we can measure the deflection of the electron by taking a difference signal between opposite segments. While quadrant detectors are a simple method to implement DPC imaging, the advantages of using a pixelated detector for a 4D-STEM DPC measurement is that we have isotropic sensitivity to both the magnitude and direction of probe shifts. Quadrant-detector DPC measurements will be optimized for a limited range of shift magnitudes and will have a CTF that depends on the direction of the potential gradients relative to the quadrant orientations.

In 4D-STEM, we measure the average deflection of the probe by computing the first moment of all DP intensities in both the x and y directions, giving a vector field over all probe positions $\mu(\mathbf{R})$. We evaluate this moment as

$$\mu(\mathbf{R}) = \left(\sum_{\mathbf{k}} I(\mathbf{k}, \mathbf{R}) k_x, \sum_{\mathbf{k}} I(\mathbf{k}, \mathbf{R}) k_y \right) / \sum_{\mathbf{k}} I(\mathbf{k}, \mathbf{R}). \quad (17.37)$$

The first moment is also known as the center-of-mass, hence the name CoM-DPC for this technique (originally listed in ■ Table 17.1). Additionally, in 4D-STEM we can apply filters in diffraction space, for example measuring only rigid shifts of the 000 beam to strongly weight our DPC measurement towards long-range, slowly varying fields.

In a DPC measurement, we typically want to reconstruct the projected potential of the specimen. We can perform this reconstruction by noting that each probe position records the vector gradient (or derivative) of the specimen's projected potential. The inverse operation for a derivative is antidifferentiation, or integration. We can robustly implement this operator in Fourier space using the expression.

$$\varphi(\mathbf{R}) = -\frac{i}{4} \mathcal{F}_{\mathbf{k} \rightarrow \mathbf{r}} \left\{ \frac{\mathbf{k} \cdot \mathcal{F}_{\mathbf{r} \rightarrow \mathbf{k}} \{ \mu(\mathbf{R}) \}}{|\mathbf{k}|^2} \right\} \text{ where } |\mathbf{k}| > 0. \quad (17.38)$$

where $\varphi(\mathbf{R})$ is the specimen's phase shift, proportional to its projected potential. If we are to use the above operator numerically, we need to avoid dividing by zero at the origin pixel in Fourier space. The above procedure will produce a measurement of the specimen potential approximately convolved by the probe's intensity distribution. Because this reconstruction method uses a FT, we also recommend padding the reconstruction

space and applying Eq. 17.38 iteratively to prevent wraparound artifacts. We can also use computational-imaging methods to reduce residual aberrations such as defocus from a DPC measurement, but generally in this case we find it's more efficient to use ptychography to fully deconvolve the probe wavefunction from the specimen's object wave.

17.6.2 Direct Ptychography

You've already come across ptychography in diffractive imaging in ► Chap. 9 and STEM imaging in ► Chap. 16. So, you should know that ptychography is the name of a family of computational-imaging methods that typically uses the coherent interference of DPs measured from overlapping converged probes to reconstruct the complex wavefunctions of both the probe and the object wave. Doing this allows the probe to be fully deconvolved from the object, including any aberrations present. We can divide ptychographic reconstruction algorithms into two groups: direct and iterative. Here we'll discuss a few of the direct methods, where we perform a single calculation to estimate the object wave. If we assume our specimen is thin enough to be considered as a 2D projected transmission function $t(\mathbf{r})$, the DPs generated by a probe $\psi_0(\mathbf{r}, \mathbf{R})$ at position $\mathbf{R}$ will have an intensity given by

$$I(\mathbf{k}, \mathbf{R}) = |\mathcal{F}_{\mathbf{k} \rightarrow \mathbf{r}} \{ \psi_0(\mathbf{r}, \mathbf{R}) t(\mathbf{r}) \}|^2. \quad (17.39)$$

If we take a 2D FT with respect to the probe positions $\mathbf{R}$, giving $I(\mathbf{k}, \mathbf{K}) = \mathcal{F}_{\mathbf{r} \rightarrow \mathbf{k}} \{ I(\mathbf{k}, \mathbf{R}) \}$, and assume the specimen is a weak-phase object ($t(\mathbf{r}) \approx 1 + i\varphi(\mathbf{r})$ where $|\varphi(\mathbf{r})| \ll 1$), the resulting intensity can be written as

$$I(\mathbf{k}, \mathbf{K}) = |\Psi_0(\mathbf{k})|^2 \delta(\mathbf{k}) + \Psi_0(\mathbf{k}) \Psi_0^*(\mathbf{k} + \mathbf{K}) T^*(-\mathbf{k}) + \Psi_0^*(\mathbf{k}) \Psi_0(\mathbf{k} + \mathbf{K}) T(\mathbf{k}), \quad (17.40)$$

where $T(\mathbf{k}) = \mathcal{F}_{\mathbf{r} \rightarrow \mathbf{k}} \{ t(\mathbf{r}) \}$ and $*$ is the complex conjugate operator. This expression contains the transmission function in Fourier space $T(\mathbf{k})$ where each spatial frequency has been shifted into the so-called 'double overlap' region, where two copies of the initial probe $\Psi_0(\mathbf{k})$ have been shifted by $-\mathbf{k}$ and $\mathbf{k}$ in addition to the unshifted probe at the origin. We can in this way estimate the coefficient of the specimen-transmission function at wavevector $\mathbf{k}$ by constructing a mask where we shift the probe-aperture function to each of these frequencies, keeping only the double overlap region and excluding the triple overlap region. By summing up these masked regions for each spatial frequency and then performing an inverse FT, we can recover the object wavefunction $t(\mathbf{r})$. Because we only need to use one of the 'sidebands' of the 4D dataset, this method is named single-sideband (SSB) reconstruction.

A direct ptychographic reconstruction method that is more accurate than SSB is the Wigner-distribution deconvolution (WDD) method. The first step in this algorithm is to estimate of the probe wavefunction, and then we compute the Wigner distribution of the probe. We can then perform a deconvolution to estimate the Wigner distribution of the object, and finally recover an estimate of the object wavefunction. The WDD method enjoys several advantages over SSB, including lifting the weak-phase approximation and allowing for super-resolution imaging. However, WDD is computationally quite demanding and has a high computer memory requirement (as does SSB), at least when the initial WDD formalism is used. So, it's not surprising that there have been several methods proposed to implement WDD in a more efficient manner.

Most ptychography experiments do not use SSB or WDD. This is because iterative ptychographic methods can more easily apply constraints and reduce artifacts due to errors in probe aberration or shape estimates and are significantly easier to parallelize. However, both the SSB and WDD methods still find applications in time-sensitive experiments such as live ptychographic reconstructions. SSB and WDD can in some cases produce more robust reconstructions for experiments with a very low electron fluence. SSB is also a very efficient way to estimate the aberration coefficients quickly and accurately, from 4D-STEM scans of highly periodic specimens.

17.6.3 Iterative Ptychography

When we use iterative ptychography algorithms we seek to model the probe and object wavefunctions by performing a forward simulation, comparing the resulting diffraction intensities to the measured DPs, and then back propagating the intensity difference signals into updates for the probe and object waves. You'll find there are many published algorithms for iterative ptychography, but here we'll cover only a few gradient-descent algorithms to give a flavor for how you can perform these reconstructions.

After collecting a 4D-STEM dataset with a convergence semi-angle, probe step size, and defocus such that the STEM probes overlap on the specimen surface, we next shift these experimental-intensity measurements $I_{\text{exp}}(\mathbf{k}, \mathbf{R})$ to the origin of diffraction space and compute the experimental amplitudes.

$$A_{\text{exp}}(\mathbf{k}, \mathbf{R}) = \sqrt{I_{\text{exp}}(\mathbf{k}, \mathbf{R})}. \quad (17.41)$$

We then construct an initial guess for the probe at the origin $\Psi_0(\mathbf{k})$ including aberrations such as defocus, and

an initial guess for the object in real space $t(\mathbf{r}) = 1$. We model the forward scattering to produce a 4D dataset containing the complex wavefunctions $\Psi(\mathbf{k}, \mathbf{R})$ at each probe position $\mathbf{R}$ as

$$\Psi(\mathbf{k}, \mathbf{R}) = \mathcal{F}_{\mathbf{r} \rightarrow \mathbf{k}} \{ \mathcal{F}_{\mathbf{k} \rightarrow \mathbf{r}} \{ \Psi_0(\mathbf{k}) \exp(-2\pi i \mathbf{k} \cdot \mathbf{R}) \} t(\mathbf{r}) \}. \quad (17.42)$$

We next calculate an update for the object wave which will reduce the difference between our forward calculation and the experimental measurements. We could compute an update function $\Delta t(\mathbf{r})$ by applying gradient descent to the above expression and solving for the difference in $t(\mathbf{r})$, giving

$$\Delta t(\mathbf{r}) = \frac{1}{S} \sum_S \frac{\mathcal{F}_{\mathbf{k} \rightarrow \mathbf{r}} \{ [A_{\text{exp}}(\mathbf{k}, \mathbf{R}) - |\Psi(\mathbf{k}, \mathbf{R})|] \exp[i\angle\Psi(\mathbf{k}, \mathbf{R})] \}}{\psi_0(\mathbf{r}, \mathbf{R})}, \quad (17.43)$$

where Σ_S represents an average of some subset S of the probe positions, $\psi_0(\mathbf{r}, \mathbf{R}) = \mathcal{F}_{\mathbf{k} \rightarrow \mathbf{r}} \{ \Psi_0(\mathbf{k}) \exp(-2\pi i \mathbf{k} \cdot \mathbf{R}) \}$ are the initial complex probes at positions $\mathbf{R}$, and $\angle\Psi(\mathbf{k}, \mathbf{R})$ is the complex argument (or angle) of the calculated wavefunctions. However, this update step is numerically unstable because the probe amplitudes $\psi_0(\mathbf{r}, \mathbf{R})$ in the denominator go to zero quickly away from their center coordinate $\mathbf{R}$. To correct for this imbalance, we can weight the update steps by the probe intensities $|\psi_0(\mathbf{r}, \mathbf{R})|^2$. Multiplying this term into the above expression as a weighting term for each position $\mathbf{R}$ and then divided by the total weight gives

$$\Delta t(\mathbf{r}) = \frac{\sum_S \mathcal{F}_{\mathbf{k} \rightarrow \mathbf{r}} \{ [A_{\text{exp}}(\mathbf{k}, \mathbf{R}) - |\Psi(\mathbf{k}, \mathbf{R})|] \exp[i\angle\Psi(\mathbf{k}, \mathbf{R})] \} \psi_0^*(\mathbf{r}, \mathbf{R})}{\sum_S |\psi_0(\mathbf{r}, \mathbf{R})|^2}, \quad (17.44)$$

where $\psi_0^*(\mathbf{r}, \mathbf{R})$ is the complex conjugate of the initial probes in real space. A similar equation is used to update the probe wavefunction, either at the same time or in alternating iterations. In practice, we also multiply the update step by some pre-factor <1 for numerical stability, and we apply regularization to the probe and object wavefunctions. If the subset S is equal to a single probe at a time, the algorithm becomes a stochastic gradient descent nearly identical to the well-known extended ePIE method. If the subset S is all probes, the algorithm is no longer stochastic and reduces to just gradient descent. Typically, smaller subsets S will improve convergence of the probe wavefunction but will increase calculation times and may worsen convergence of the object wavefunction. The reconstruction workflow for iterative ptychography is shown in  Fig. 17.6.

There are many extensions to these ptychographic reconstruction methods. The most common is to use multiple object slices to handle multiple scattering, known as multi-slice ptychography. Another is to use multiple incoherent waves for the probe, the object, or

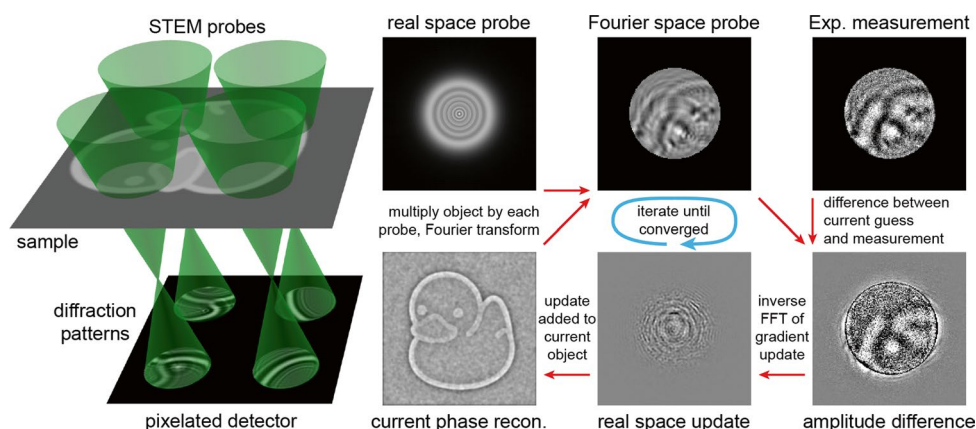


Fig. 17.6 4D-STEM iterative ptychography workflow showing defocused probes. First, we record a 4D-STEM dataset with sufficiently large defocus and small enough step size to guarantee enough probe overlap (redundancy in the data). To reconstruct the object, we iteratively forward propagate each probe, compare it to each experimentally measured intensity, and compute a gradient update step through back propagation for both the object and the probe. We iterate through all probe positions repeatedly, until convergence

both, a method known as mixed state ptychography. We can also combine multiple experimental measurements with different probe wavefunctions, for example measuring 4D-STEM datasets at different defocus values to reconstruct the full scattering matrix of the object. The probe wavefunction may also be modified, for example by using phase plates to add phase diversity to the measurements. So, there's lots of room for you to practice current (and develop future) computer simulations!

chanical straining of the specimen, electrical biasing, and gas or liquid flow. More information on in situ TEM experiments is given in ► Chap. 3 in this volume. In-situ 5D-STEM has been used for various NBED experiments to measure the time evolution of material properties such as strain, orientation, and crystalline ordering or disorder. We can also directly measure the time dependence of fluctuations from disordered materials, as an extension to FEM experiments.

17.7 Beyond 4D

So it should be very clear by now that 4D-STEM represents a significant increase in STEM data collection, enabled by fast DEDs. However, these detectors are still improving and becoming more ubiquitous in STEM, and the computational power available to all researchers increases each year. So, you'll see in the literature that many microscopists are expanding their 4D-STEM studies by adding additional measurements. These include simultaneous measurement channels such as HAADF or XEDS characterization which add data but do not increase the dimensionality of the experiment, and sequential 4D-STEM measurements adding additional dimensions such as time, specimen tilt in one or more directions, and others. Here, we briefly summarize some of the possible extensions to 4D-STEM that you can find in the literature.

17.7.1 In-Situ 4D-STEM

The most straightforward 5D-STEM experiment is a time series of 4D-STEM scans, typically with external stimuli applied to the specimen which may vary over time. These stimuli can include heating and me-

17.7.2 Defocus or Other Aberrations Series 4D-STEM

Some phase-contrast 4D-STEM methods such as iterative multislice ptychography or scattering-matrix reconstructions are limited in how accurately they can reconstruct thicker specimens. This is because sampling increasingly finely does not produce different enough measurements to shed more light on specimen structure. One method to gather such information is to perform multiple 4D-STEM scans, changing the probe wavefunction between them. The most straightforward probe aberration to introduce is defocus, i.e. performing a depth-sectioning 5D-STEM experiment. The electron probe can be modulated by any aberrations, or a phase plate can be used to change the wavefunction between scans. Adding this 'phase diversity' has been shown to improve ptychographic reconstructions, though more 3D information can be recovered by tilting the specimen between scans than by modulating aberrations such as defocus.

17.7.3 Tilt Series 4D-STEM for Tomography

As discussed above, 4D-STEM enables phase-contrast STEM experiments using DPC or ptychography.

These experiments can be combined with specimen tilting to reconstruct the specimen structure in 3D. There's more information on tomography in ► Chap. 19. These 5D-STEM experiments can be computationally expensive, since they require many ptychographic reconstructions for each experiment, even before the data is reconstructed in 3D. This alignment procedure may also be challenging, because ptychographic reconstructions will produce contrast from virtually all specimen structures, including from carbon substrates or contamination. One method that can help you with alignment is using fiducial markers, typically heavy-element nanoparticles. However, these fiducials are very challenging to use at atomic resolution. Recently, researchers have explored the possibility of performing joint ptychography-tomography reconstructions, where they perform iterative multislice ptychography on the full 3D volume as it is being reconstructed. While you can see that this combination is even more computationally challenging, it's expected to produce the highest-quality reconstructions as it uses all available experimental information. As an aside, we can also use 4D-STEM tomography to reconstruct the magnetic structure of a specimen.

17.8 Complex 4D-STEM Experiments

In each of the above sections we've described 4D-STEM experiments using relatively standard TEM configurations, except perhaps for using amplitude and phase plates in the probe-forming aperture. However, there are many lenses in a modern, aberration-corrected TEM, which we can configure in a multitude of different ways. So now we'll describe some of these configurations which are potentially useful for 4D-STEM experiments.

17.8.1 Real Space and Confocal Imaging 4D-STEM

Our typical STEM and 4D-STEM experiments perform measurements in the far field after the specimen, i.e. in a diffraction plane. However, we can, instead, use the lenses after the specimen to refocus the beam, forming a conjugate image plane at the detector. With this configuration we are imaging the real-space distribution of the electron probe at the exit surface of our specimen, and we can push this method all the way to atomic resolution. When we combine this configuration is combined with a large α_{probe} and scanning, it is referred to as scanning confocal electron microscopy (SCEM). With SCEM we can achieve significantly better depth resolution than with other STEM methods,

as electrons scattered from different heights within a specimen can be spatially separated at the detector. At atomic resolution, SCEM has also directly imaged electron channeling along atomic columns.

17.8.2 4D Scanning Confocal Electron Diffraction

In a typical NBED experiment, we place the specimen at the electron-beam crossover, where the probe is close to being in focus, and therefore we measure DPs in the far field. However, we can instead move the specimen far above or below the crossover, where the intersection of the beam and specimen have a circular profile. This has the advantage of minimizing potential damage in beam-sensitive specimens due to the long tails of probes with an Airy-disk function shape. As in 4D-SCEM, we can measure DPs by using additional lenses below the specimen to refocus the beam. Then we produce diffraction 'spots' that are blurrier than a typical NBED experiment due to being modulated by the thickness of the specimen. This method is referred to as a 4-D scanning confocal electron diffraction (4D-SCED). These experiments require that you carry out a very careful alignment of the optical axis and the de-scan coils of the beam and the patterns may suffer from additional aberrations. Nevertheless, the increased complexity of the 4D-SCED method may be a worthwhile tradeoff for very beam-sensitive specimens.

17.8.3 Energy-Resolved 4D-STEM

Almost every STEM instrument includes an EELS (if yours doesn't you should get one asap!) As you know from W&C Chaps 37–40, EELS has a multitude of uses. Here we'll briefly describe the combination of EELS and 4D-STEM by measuring energy-resolved DPs or equivalently momentum-resolved energy measurements. With an EELS, we can measure a full DP after applying an energy slit. By repeating the scan with different energy values for the slit, we can measure a 5D-STEM dataset where the added dimension corresponds to energy. This approach can be used to study DPs, for example from specific plasmon or core-loss energy transitions in our specimen. We can alternatively change the spectrometer optics such that energy is resolved along one direction and momentum along the orthogonal direction. This configuration has much better energy resolution and we can use it to study electronic band structures, *inter alia*. You can also repeat the measurements with various specimen tilts to recover the missing momentum axis, although these datasets are really challenging to align.

17.9 4D-STEM Analysis Software

If you're only used to working with 2D images and DPs in STEM and TEM, you may find working with 4D-STEM data difficult. This is because, rather than working with a small number of images each with a size of 1–10 MB, 4D-STEM datasets can potentially be extremely large, from GB to TB. If each DP is only 250 x 250 pixels, then saving the data with 2 bytes per pixel will require only 250 kilobytes per image. However, if we record the same number of probe positions as in a typical STEM experiment, for example 1000 x 1000 positions, the total raw dataset will require 125 GB of storage, far larger than the typical memory of most desktop computers and laptops. Even if we switch to a sparse, list-based file format, as we described in the event-driven detectors section above, the file sizes can still be very large. If we record 2500 electrons per image (2 bytes each) for 1000 x 1000 probe positions, the resulting datasets will be 5 GB. This file size is more comparable to 3D STEM experiments such as in situ movies, tilt-series tomographic reconstructions or spectroscopic XEDS or EELS experiments, but with modern DEDs we could record many of these in a single microscope session. Additionally, many 4D-STEM analysis algorithms require us to 'inflate' these sparse images into full images which may increase the computer memory requirements still further.

17.9.1 Open Source 4D-STEM Analysis Codes

Disclosure

The author of this particular Chapter is one of developers of py4DSTEM.

Because this field is still relatively new, most 4D-STEM analysis methods have been developed by microscopists as open-source software. These range from relatively small single-experiment analysis methods to large software toolkits or even analysis ecosystems. Currently the three largest 4D-STEM analysis codes are py4DSTEM,

liberTEM, and various HyperSpy sub-packages such as Orix and pyxem. It is no accident that all these tools are written in the Python language, which is the most widely used programming language for scientific-analysis methods. If you're familiar with Python you'll know that it allows you to easily mix and match the various codes and methods, and integrate methods from other fields. For example, many of the tools developed by the X-ray and optical diffraction and imaging communities can be used for 4D-STEM analysis, and Python tomographic reconstruction packages can be used to change many 4D-STEM analyses into 3D. We would also be remiss not to mention ML methods, which have, in large part, displaced conventional image-processing methods. Python is also the language of choice for these methods, of which the most popular are pyTorch, TensorFlow, and Jax. ■ Table 17.2 lists some open-source analysis software.

17.9.2 Closed-Source 4D-STEM Analysis Codes

TEM hardware vendors and detector manufacturers primarily still use closed-source, highly fragmented, software programs to run the experiments and perform analysis. This is particularly common with fast DEDs used for 4D-STEM experiments, which may be run from a separate computer and will probably save the data into an undocumented, closed-file format. Such software makes it very difficult to collect all the required meta-data for your 4D-STEM experiment, and thus these experiments often require significant calibration during analysis. In ■ Table 17.3 we've summarized some of the vendor-provided 4D-STEM analysis software.

With vendor software, we often either have no choice but to use their tools, or sometimes they can provide a lot of value due to their close integration with the microscope. There are non-vendor, closed-source, analysis-software codes developed by scientists and scientific companies, which we'll not outline here. We recommend against using such closed-source analysis codes, since they cannot be validated or checked for accuracy, and are often difficult to reproduce results from.

■ Table 17.2 A non-exhaustive list of open source 4D-STEM analysis codes

Name	Notes
py4DSTEM	General code with a focus on versatility and usability
Pyxem	Part of the HyperSpy family of open-source codes, focused on NBED
LiberTEM	General code with a focus on high throughput distributed processing

Table 17.3 A non-exhaustive list of vendor-provided 4D-STEM analysis codes

Name	Notes
DigitalMicrograph	A general platform for experiments, supports some analysis workflows. Also known as the Gatan Microscopy Suite
NION Swift	GUI python software for microscope control, supports some analysis workflows
NOVENA	Detector control software, supports some analysis workflows

17.9.3 Simulations of 4D-STEM Experiments

Much like with conventional TEM experiments, we often wish to validate or interpret our 4D-STEM experiments using simulations. We can also use simulations to test new analysis methods on datasets where the ground truth is known, something which is not possible with real-world experiments (if we know the ground truth why are we doing the experiment...?). As with software analysis of 4D-STEM experiments, the primary challenge we face with 4D-STEM simulations is the large dataset sizes and long simulation times often required. There are four primary methods to simulate 4D-STEM experiments, which we explain in detail below. [Table 17.4](#) lists some open-source codes which can be used to simulate 4D-STEM experiments.

As you've learned in W&C [Chap. 12](#), the simplest method to simulate diffraction from a crystalline specimen is kinematical scattering using the Ewald-sphere construction. We make the (extremely strong) simplifying assumption that our specimen will only scatter electrons into a small number of scattering vectors. We make an additional assumption that our specimen is thin enough that the scattering is purely kinematical. As you know, this means each electron will scatter only once from the initial incident-beam direction into a given Bragg peak, and that the beam intensity is proportional to the crystal's structure factor for that beam (modulated as you know by the shape factor of the specimen). If we use this method for simulation of a STEM experiment where we have multiple, incident-beam directions (all the angles less than the maximum scattering angle of our

probe-forming aperture), we will assume they are all linearly independent from each other, or that the intensity of each Bragg disk is constant (i.e., flat disks in diffraction space). This method requires that the specimen is periodic, but because it is very computationally efficient it is often used to compute lookup tables for diffraction of crystalline specimen orientations and phases.

The second method to simulate diffraction from a crystalline specimen is the Bloch-wave construction (check out the daunting [Chap. 14](#) in W&C). This construction allows electrons to scatter multiple times between different beams. We keep the assumption that only a small number of scattering vectors are allowed (diffraction-space sparsity), but in addition to including multiple or dynamical scattering, we can also include some approximate inelastic scattering by including absorption. The Bloch-wave method works by first constructing a structure matrix for our specimen, and then transforming it into a scattering matrix. We 'solve' the scattering matrix by computing its eigenvectors, each of which represents an electron wave which is periodic along the beam direction, with the periodicity length given by the associated eigenvalue. Because these eigenstates form a complete basis, we can compute their representation for any incident-probe wavefunction, and then use a matrix multiplication to compute the exit wavefunction. The Bloch-wave method is highly efficient for small periodic unit cells, but scales very poorly with the number of scattering pathways between beams and thus we can only use it for small periodic unit cells. Because this construction computes the eigenstates of the electron wave inside our specimen, we can use it to calculate diffraction from

Table 17.4 A non-exhaustive list of open source 4D-STEM simulation codes

Name	Notes
abTEM	A general purpose python code for STEM and TEM simulation
Prismatic	GUI and c++ software for STEM imaging and diffraction simulations
μ STEM	FORTTRAN code for STEM and TEM simulation
py_multislice	A general purpose python code for STEM and TEM simulation
Dr. Probe	General purpose simulation code with a GUI
MULTEM	A general purpose c++ code for STEM and TEM simulation
STEMsalabim	Parallelized simulation routines

many different specimen thicknesses simultaneously. This property makes it very useful for computing lookup tables for specimen thickness, in addition to a specimen's phase and orientation.

The third and most common method for simulating 4D-STEM experiments is the multislice algorithm (as introduced in Chap. 30 in W&C). In the multislice method, we first split up our specimen into many thin slices along the beam direction and compute the atomic potentials using a parameterization method or a first principles method such as density functional theory (DFT). We then solve the Schrodinger equation for a thin slice, by splitting it into two operators which we apply sequentially: transmission through the slice as a phase grating, and then propagation to the next slice. By alternating these two operations until the beam has passed through all specimen slices, we can compute the complex wavefunction of our specimen for virtually any incident probe and detector configuration we wish. The multislice method is highly accurate, flexible, and easy to implement which makes it the most common method for simulating 4D-STEM experiments, especially those which do not consist of small periodic cells.

However, the multislice algorithm can be quite inefficient for STEM simulations, as it requires a new

simulation for each incident-probe wavefunction (which includes different positions on the specimen surface). If the STEM probe is much smaller than the cell being simulated, we could crop around each probe to reduce the simulation time by a factor of f^2 , where f is the relative cropping factor in x and y . However, the fourth method for simulating 4D-STEM experiments can reduce calculation times by orders of magnitude. This method is known as the plane-wave reciprocal-space interpolated scattering matrix (PRISM). It works by using the multislice method to compute plane-wave simulations at different incident angles, which become rows of a scattering matrix. In much the same way as the Bloch-wave method, this scattering matrix can be used to compute the output wavefunction for any STEM probe initial condition. We can speed up the calculation by using Fourier interpolation to crop around the probe, which speeds up the scattering matrix calculation by a factor of f^2 , and the evaluation of the output STEM probes by f^4 . If $f = 1$, the PRISM method is formally identical to the multislice method, though usually slower. However, for $f \geq 2$, PRISM is usually faster than multislice. The primary limitation of PRISM is that we must take care to ensure that the STEM probe size never exceeds the cropping-box dimensions, or else the accuracy will be reduced (especially for high-angle scattering).

Chapter Summary

This chapter summarizes the rapidly developing family of 4D-STEM experiments. These experiments include mapping phase, orientation and strain in crystalline materials, pair-distribution function mapping, FEM, angular correlations, and strain mapping of disordered specimens. We have covered the experimental parameters and provided a brief mathematical description of the analysis methods for each of these experimental families. The most important microscope parameters for 4D-STEM are:

- α_{probe} the probe convergence semi-angle.
- Aberrations, amplitude plates and phase plates in the probe-forming aperture.
- Step size between adjacent probe position, scan patterns.
- Detector characteristics, sampling, and maximum value of scattering angles.

One of the most important uses for 4D-STEM is phase-contrast imaging, especially for weakly scattering materials and light elements. The most important phase

contrast diffractive imaging methods in 4D-STEM are DFC, and direct or iterative ptychography. We have also covered more complex 4D-STEM experiments, including real-space and confocal-imaging modes, and experiments combining 4D-STEM with spectroscopy or tomographic tilt series.

Performing complex 4D-STEM analyses almost always requires the use of computational methods. We strongly recommend that you learn programming and mathematical microscopy – we promise that these skills will transfer to many other aspects of electron microscopy and science in general!

We have also covered the software methods which can be used for 4D-STEM analysis, and simulation codes which are 4D-STEM-ready. We have discussed the importance of open-source software for 4D-STEM analysis, and here we re-emphasize the importance of *including all raw data, metadata, and analysis steps in your publications*.

People

4D-STEM is a very large umbrella, encompassing the work of many TEM researchers. John Cowley was one of the first to imagine the possibilities of spatially-resolved diffraction experiments, like many other microscopy techniques. Other early 4D-STEM pioneers include Peter Nellist, John Rodenburg, Nestor Zaluzec, Paul Voyles, Kazuo Ishizuka, Jian-Min Zuo, Stephen Pennycook, and many others. However, as described above, the popularity of 4D-STEM is directly linked to the availability of high-speed DEDs. The use of hybrid PAD detectors for 4D-STEM applications was driven by David Muller, Sol Gruner, and their colleagues. Widely used APS detectors in 4D-STEM were developed by Peter Denes, Xuong Nguyen-Huu, Stuart Kleinfelder and their colleagues. Several other detector developments happened in parallel, including the adaptation of CERN-developed detectors for 4D-STEM experiments.

The most important story of 4D-STEM development however has been the development of analysis software for these experiments. HyperSPY was initially created by Francisco De La Peña, and contributions important for 4D-STEM analysis have been added by Philip Crout, Niels Cautaerts, Duncan Johnstone, Carter Francis, Magnus Nord, and Håkon Wiik Ånes. LibSTEM was initially created by Alexander Clausen and Dieter Weber. py4DSTEM was initially created by Benjamin Savitzky and Colin Ophus, and major contributions been added by Steven Zeltmann, Stephanie Ribet, Georgios Varnavides, and Alexander Rakowski. Each of these open-source codes has many other contributors, as is the nature of open-source projects.

(Personal note from Colin Ophus). Let me briefly describe how I coined the name ‘4D-STEM’. While working with early high speed DEDs at the National Center for Electron Microscopy (NCEM) in Berkeley, we tried many different early 4D-STEM experiments with a wide range microscope parameters. These included nearly parallel illumination with convergence semi-angle of 0.1 mrad up to coherent imaging with a semi-angle of 30 mrad, covering about two and half orders of magnitude. I wanted to describe these techniques with a blanket term, but none of the existing names in the literature such as NBED or SEND or ptychography could encompass them all. These experiments did have a lot in common however: they were all STEM-based, produced four-dimensional datasets, and were very data-intensive and computationally challenging to analyze, especially back when hard drives were much smaller and computers significantly slower. Because the large dimensionality of the data was at the

forefront of my mind, I referred to these experiments internally as ‘4D-STEM’. After a vendor joked that they were going to trademark the name, I started using the term externally as well, in my talks and papers. I believe that in addition to being a useful descriptor, the name ‘4D-STEM’ caught on because it is catchy and fun to say.

Self-assessment Questions

- Q17.1 What are the advantages of using a pixelated detector over a monolithic detector, for a STEM experiment?
- Q17.2 What are the two main types of detectors used for 4D-STEM experiments? Which one has a higher dynamic range, and which has more pixels?
- Q17.3 How will an increase of α_{probe} affect the probe size on the specimen? How will increasing the probe defocus from zero affect its size? What about decreasing the defocus?
- Q17.4 Can we form multiple STEM probes with an amplitude plate? What about using a phase plate?
- Q17.5 Will specimen drift be better or worse if we scan the STEM probe at a higher speed?
- Q17.6 Why would we use PED?
- Q17.7 Can we measure the 3D orientation of a crystal from a single 2D DP?
- Q17.8 If the atoms of a specimen are pushed closer together, do the diffraction spots move closer together or further apart?
- Q17.9 What are the required experimental parameters to accurately measure the pair distribution function of an amorphous material?
- Q17.10 What property of the DPs do we measure in a FEM experiment?
- Q17.11 What happens to the amorphous halo in the DP of an amorphous material if we put the specimen in tension (i.e. move the atoms further apart in one direction)?
- Q17.12 Do phase retrieval STEM methods determine the crystalline phase of the specimen?
- Q17.13 What is the center-of-mass of a DP? What specimen property do changes in the center-of-mass correspond to?
- Q17.14 What are the two most common algorithms employed for direct ptychography phase retrieval?
- Q17.15 Should iterative ptychography phase retrieval methods be classified as gradient descent or stochastic gradient descent?
- Q17.16 What are some extra dimensions we could add to 4D-STEM to perform a 5D-STEM experiment?

Q17.17 What does 4D-SCED stand for, and how is it different from a NBED 4D-STEM measurement?

Q17.18 what are the names of three open-source software packages which could be used for 4D-STEM analysis?

Q17.19 What is the most common algorithm used to simulate a 4D-STEM experiment?

Q17.20 What are some limitations of kinematical diffraction simulations? What about Bloch-wave simulations?

Text-Specific Questions

- T17.1 Using the approximation given in Eq. 17.4, derive the FWHM of a STEM probe equal as a function of the α_{probe} and wavelength.
- T17.2 If we are using a STEM probe with $\alpha_{\text{probe}} = 20$ mrad and defocus $C_1 = 100$ Angstroms, calculate the required probe step size to give 80% overlap between adjacent probes.
- T17.3 Estimate how much a specimen's surface would need to be tilted away from the beam direction to increase the excitation error by 1%.
- T17.4 Write down the strain transformation matrix for strain values of $\varepsilon_{xx}=1\%$, $\varepsilon_{yy}=0\%$, and $\varepsilon_{xy}=-1\%$.
- T17.5 The PDF given in Eq. 17.25 asymptotically approaches 1, i.e. a constant density of bonds. Write down the factor we would need to multiply this expression by to give the atomic-probability density, accounting for the size of each spherical shell.
- T17.6 In a FEM experiment, if an amorphous ring has a variance of 500 electron counts squared and a mean value of 100 counts, what is the normalized variance equal to?
- T17.7 Using Eq. 17.31, derive the angular cross correlation for a ring with intensity equal to $I = \sin^2(\phi)$.
- T17.8 The eccentricity of an ellipse is given by the expression $e = \sqrt{1 - C^2/A^2}$. Calculate the eccentricity of the amorphous halo in a DP from a material with strains equal to $\varepsilon_{xx}=1\%$, $\varepsilon_{yy}=0\%$, and $\varepsilon_{xy}=-1\%$.
- T17.9 Derive the slope of a complex plane wave which deflects a STEM probe 1% of its convergence angle in diffraction space, in terms of its convergence angle and wavelength.
- T17.10 Derive a constant approximation to the denominator of Eq. 17.44.

References

General References

Bustillo KC, Zeltmann SE, Chen M, et al. (2021) 4D-STEM of beam-sensitive materials, *Accounts Chem Res* 54 2543-2551

Liu JJ (2021). Advances and applications of atomic-resolution scanning transmission electron microscopy. *Microsc Microanal* 27 943-995

Ophus C (2019) Four-dimensional scanning transmission electron microscopy (4D-STEM): from scanning nanodiffraction to ptychography and beyond. *Microsc Microanal* 25 563-582

Ophus C (2023) Quantitative scanning transmission electron microscopy for materials science: imaging, diffraction, spectroscopy, and tomography, *Annu Rev Mater Res* 53 105-141

Xie Y, Wang J, Savitzky BH, Chen Z, Wang Y, Betzler S, Bustillo K, Persson K, Cui Y, Wang L-W, Ophus C, Ercius P, Zheng H (2023) Spatially resolved structural order in low-temperature liquid electrolyte. *Sci Adv* 9(2). ► <https://doi.org/10.1126/sciadv.adc9721>

Zuo JM, Spence JCH (2013) *Electron microdiffraction*, Springer Science and Business Media.

Detectors, Phase and Amplitude Plates, Precession

McMullan G, Faruqi A, Clare D et al (2014) Comparison of optimal performance at 300 keV of three DEDs for use in low dose electron microscopy. *Ultramicroscopy* 147:156–163

Tate MW, Purohit P, Chamberlain D, et al. (2016) High dynamic range pixel array detector for scanning transmission electron microscopy. *Microsc Microanal* 22:237–249

Phase and Amplitude Plates

Harvey TR, Yasin FS, Chess JJ, et al. (2018). Interpretable and efficient interferometric contrast in scanning transmission electron microscopy with a diffraction-grating beam splitter. *Phys Rev Appl* 10, 061001

Ophus C, Ciston J, Pierce J, et al. (2016) Efficient linear phase contrast in scanning transmission electron microscopy with matched illumination and detector interferometry. *Nature Communications* 7 10719

Verbeeck J, Béch   A, M  ller-Caspary K, et al. (2018) Demonstration of a 2×2 programmable phase plate for electrons. *Ultramicroscopy* 190 58-65

Zeltmann SE, Muller A, Bustillo KC, et al. (2020) Patterned probes for high precision 4D-STEM Bragg measurements. *Ultramicroscopy* 209 112890

Precession Electron Diffraction

Midgley PA, Eggeman AS (2015) Precession electron diffraction—a topical review. *IUCrJ* 2(1) 126–136

Vincent R, Midgley P (1994) Double conical beam-rocking system for measurement of integrated electron diffraction intensities. *Ultramicroscopy* 53(3) 271–282

Nanobeam Diffraction of Crystalline Materials

Brunetti G, Robert D, Bayle-Guillemaud P, et al. (2011) Confirmation of the domino-cascade model by LiFePO₄/FePO₄ precession electron diffraction. *Chem Mater* 23(20) 4515–4524

Cooper D, Denneulin T, Bernier N, et al. (2016) Strain mapping of semiconductor specimens with nm-scale resolution in a transmission electron microscope. *Micron* 80 145–165

Morawiec A, Bouzy E, Paul H, et al. (2014) Orientation precision of TEM-based orientation mapping techniques. *Ultramicroscopy* 136 107–118

Pekin TC, Gammer C, Ciston J, et al. (2017) Optimizing disk registration algorithms for nanobeam electron diffraction strain mapping. *Ultramicroscopy* 176 170–176

Nanobeam Diffraction of Disordered Materials

Gammer C, Ophus C, Pekin TC, et al. (2018) Local nanoscale strain mapping of a metallic glass during in situ testing. *Appl Phys Lett* 112(17) 171905

Liu ACY, Neish MJ, Stokol G, et al. (2013) Systematic mapping of icosahedral short-range order in a melt-spun Zr₃₆Cu₆₄ metallic glass. *Phys Rev Lett* 110 205505

Mu K, Mazilkin A, Sprau C, et al. (2019) Mapping structure and morphology of amorphous organic thin films by 4D-STEM pair distribution function analysis. *Microscopy* 68(4) 301–309

Zhang, P, Maldonis JJ, Liu, Z, et al. (2018) Spatially heterogeneous dynamics in a metallic glass forming liquid imaged by electron correlation microscopy. *Nat Commun* 9 1129

Phase Contrast STEM

Jiang Y, Chen Z, Han Y, et al. (2018) Electron ptychography of 2D materials to deep sub-Ångström resolution. *Nature* 559(7714) 343

Lubk A, Zweck J (2015) Differential phase contrast: An integral perspective. *Phys Rev A* 91(2) 023805

Maiden AM, Rodenburg JM (2009) An improved ptychographical phase retrieval algorithm for diffractive imaging. *Ultramicroscopy* 109(10) 1256–1262

Varnavides G, Ribet SM, Zeltmann SE, et al. (2023) Iterative phase retrieval algorithms for scanning transmission electron microscopy. ► [arXiv:2309.05250](https://arxiv.org/abs/2309.05250)

Complex Extensions to 4D-STEM

Etheridge J, Lazar S, Dwyer C, et al. (2011) Imaging high-energy electrons propagating in a crystal. *Physical Review Letters* 106(16) 160802

Senga R, Suenaga K, Barone P, et al. (2019) Position and momentum mapping of vibrations in graphene nanostructures. *Nature* 573(7773) 247–50

Wu M, Harreiss C, Ophus C, et al. (2022) Seeing structural evolution of organic molecular nano-crystallites using 4D scanning confocal electron diffraction (4D-SCED). *Nat Commun* 13(1) 2911

Software Analysis of 4D-STEM Data

Cautaerts N, Crout P, Ånes HW, et al. (2022) Free, flexible and fast: orientation mapping using the multi-core and GPU-accelerated template matching capabilities in the python-based open source 4D-STEM analysis toolbox Pyxem. *Ultramicroscopy* 237 113517

Clausen A, Weber D, Ruzaeva K et al (2020) LiberTEM: software platform for scalable multidimensional data processing in transmission electron microscopy. *J Open Source Software* 5(50) 1-4

Johnstone DN, Martineau BH, et al. (2020) Density-based clustering of crystal (mis) orientations and the orix Python library. *J Appl Crystallogr* 53(5) 1293–1298

Savitzky BH, Zeltmann SE, Hughes LA, et al. (2021) py4DSTEM: a software package for four-dimensional scanning transmission electron microscopy data analysis. *Microsc Microanal* 27(4) 712–743